

Unified Multi-Environment Cyber Threat Detection and Response Platform

*A dissertation submitted in partial fulfillment of the requirements
for the award of the degree of*

MASTER OF COMPUTER APPLICATIONS

By

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CERTIFICATE

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ABSTRACT

Modern computing infrastructures span complex heterogeneous environments—including on-premises Linux/Windows hosts, virtualized container clusters, cloud workloads, and multi-tenant networks. Security teams face extreme operational challenges detecting intrusions across disparate telemetry formats, handling overwhelming alert volumes, and executing timely mitigations without disrupting critical production services. Traditional SIEM solutions suffer from prohibitive licensing costs, rigid proprietary query languages, delayed batch alerting, and an absence of cryptographic auditability.

The Unified Multi-Environment Cyber Threat Detection and Response Platform addresses these vulnerabilities by establishing a modern, high-throughput, and tamper-evident Security Operations Center (SOC) framework. The platform normalizes diverse telemetry into the Elastic Common Schema (ECS) standard and executes a concurrent dual-engine detection architecture combining declarative YAML rule evaluation with rolling Z-score statistical anomaly baselines.

A multi-stage kill-chain correlation engine maps emerging threat activity against the MITRE ATT&CK framework, identifying advanced persistent threats (APTs) across initial access, execution, lateral movement, and exfiltration phases. To eliminate the operational risks of fully autonomous response, the platform enforces a Human-in-the-Loop approval gate for destructive mitigations (such as firewall IP blocks and account disablement).

Every mutating administrative event, detection alert, and remediation action is committed to an immutable SHA-256 hash-chained cryptographic audit ledger, guaranteeing mathematical tamper resistance in compliance with ISO/IEC 27001 and SOC 2 Type II audit standards. Empirical evaluation confirms sustained ingestion throughput exceeding 1,250 events per second, sub-20ms detection latency, a 96.4% True Positive Rate, and seamless real-time WebSocket state distribution.

Technologies Used: Python 3.12, FastAPI, React 18, TypeScript, Tailwind CSS, PostgreSQL 16, Redis 7.2 Streams, PyJWT, Argon2id, Docker Compose, RESTful APIs, WebSockets.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
API	Application Programming Interface
APT	Advanced Persistent Threat
Argon2id	Key Derivation Function for Password Hashing (Winner of PHC)
CORS	Cross-Origin Resource Sharing
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
DFD	Data Flow Diagram
ECS	Elastic Common Schema (Standard Log Specification)
ER	Entity-Relationship
EVTX	Windows XML Event Log File Format
GDPR	General Data Protection Regulation
HOD	Head of the Department
HTML	Hypertext Markup Language
HTTP	Hypertext Transfer Protocol
HTTPS	Hypertext Transfer Protocol Secure
IOC	Indicator of Compromise (IP, Hash, Domain)
IT	Information Technology
JSON	JavaScript Object Notation
JWT	JSON Web Token (RFC 7519)

MCA	Master of Computer Applications
MISP	Malware Information Sharing Platform & Threat Sharing
MTTR	Mean Time to Respond
NLP	Natural Language Processing
ORM	Object-Relational Mapping
OTX	AlienVault Open Threat Exchange
RBAC	Role-Based Access Control
REST	Representational State Transfer
SDLC	Software Development Life Cycle
SHA-256	Secure Hash Algorithm (256-bit Digest)
SIEM	Security Information and Event Management
SOAR	Security Orchestration, Automation, and Response
SOC	Security Operations Center
SPA	Single Page Application
SQL	Structured Query Language
SRS	Software Requirements Specification
UAT	User Acceptance Testing
UI	User Interface
UML	Unified Modeling Language
UX	User Experience
Vite	Next-Generation Frontend Tooling & Bundler
VM	Virtual Machine
WSS	WebSocket Secure Protocol
YAML	YAML Ain't Markup Language (Rule Definition Format)

CHAPTER – 1

Introduction

1.1 Background and Overview

Modern enterprise networks, academic institutions, and cloud infrastructures operate under a continuous deluge of cyber threats. Multi-environment systems—comprising heterogeneous Linux servers, Windows Active Directory domains, Amazon Web Services (AWS) cloud workloads, and distributed IoT endpoints—generate millions of operational telemetry logs daily. Any unmonitored server, misconfigured firewall port, compromised credential, or unpatched service represents an immediate entry point for malicious adversaries.

Traditionally, institutional and enterprise security operations rely on manual inspection, disparate logging utilities, or siloed tools. A system administrator reviews firewall logs in one console, inspects Windows Event Viewer on individual hosts, and relies on user reports when services fail or accounts are compromised. Such fragmented, manual workflows lack real-time correlation, produce no unified audit trail, and result in severe security backlogs, long dwell times, and catastrophic breaches. As network complexity and data volume expand, this ad-hoc model becomes untenable.

The Unified Multi-Environment Cyber Threat Detection and Response Platform addresses this critical security deficit. It is a full-stack, real-time Security Operations Center (SOC) platform engineered to ingest heterogeneous telemetry logs, normalize them into the standardized Elastic Common Schema (ECS), evaluate security events concurrently through declarative YAML rules and statistical anomaly baselines, correlate kill-chain intrusions against the MITRE ATT&CK matrix, and execute automated remediations guarded by Human-in-the-Loop review gates.

1.1.1 Cyber Threat Intelligence & Multi-Environment Landscape

Reviewing how enterprise and campus network security is currently handled reveals five recurring shortcomings that motivate the present system:

- 1. No Centralized Telemetry Record (Critical): Security logs across disparate endpoints, web servers, and cloud accounts are fragmented, making cross-platform attack tracing impossible.
- 2. Absence of Live Threat Visibility (High): Security teams lack unified single-pane visibility, remaining unaware of active brute-force or lateral movement campaigns until substantial damage occurs.

- 3. Excessive Manual Triage Overhead (High): SOC analysts are overwhelmed by thousands of disconnected alerts, spending valuable hours manually categorizing events and mapping tactics.
- 4. Redundant and False-Positive Alerts (Medium): Repetitive benign anomalies and un-correlated alert storms consume analyst bandwidth and induce severe alert fatigue.
- 5. Operational Risks in Autonomous Remediation (Critical): Uncontrolled automated blocking risks isolating mission-critical production servers due to benign spikes, demanding strict human-in-the-loop validation.

1.2 Motivation and Problem Statement

1.2.1 Core Motivation

The motivation for this project stems from direct observation of modern cybersecurity warfare. Advanced Persistent Threats (APTs) do not operate in a single domain; an intrusion may begin with phishing on an endpoint, execute via PowerShell on a Windows host, move laterally through Linux database servers, and exfiltrate data to an external cloud bucket. Detecting such multi-stage attacks requires an interconnected, high-throughput defense platform capable of parsing disparate log structures and correlating activities across time and network boundaries.

1.2.2 Problem Statement

There is no lightweight, transparent, and mathematically verifiable platform through which multi-environment security events can be ingested, normalized, correlated, and mitigated in real time without incurring exorbitant licensing fees or risking operational paralysis from unvetted automation. The problem addressed by this dissertation is therefore: how can a full-stack web-based platform, augmented with statistical anomaly engines, declarative YAML rules, and cryptographic audit ledgers, be engineered to streamline the entire threat detection and response lifecycle across heterogeneous environments in an auditable, high-performance, and cost-effective manner?

1.3 Aim and Objectives

1.3.1 Aim of the Project

To design, implement, and evaluate a production-grade, full-stack, role-based platform—the Unified Multi-Environment Cyber Threat Detection & Response Platform—that unifies security telemetry ingestion, automated dual-engine threat detection, MITRE ATT&CK correlation, human-gated response automation, and cryptographic SHA-256 audit chaining.

1.3.2 Specific Objectives

- 1. To design a relational and time-series data model capable of representing normalized events, detection rules, threat incidents, response actions, and cryptographic audit blocks.
- 2. To implement a secure, stateless JWT-based authentication and role-based access-control (RBAC) layer distinguishing Security Analysts, Incident Responders, and SOC Administrators.
- 3. To construct a multi-environment log ingestion pipeline supporting Syslog (RFC 5424), Windows EVTX XML, and AWS CloudTrail JSON into canonical ECS fields.
- 4. To build a dual-engine threat detection pipeline executing declarative YAML signature rules concurrently with rolling Z-score statistical anomaly baselines.
- 5. To implement MITRE ATT&CK multi-stage kill-chain correlation that aggregates isolated alerts sharing entity keys into unified, prioritized Security Incidents.
- 6. To engineer an automated response dispatcher with a Human-in-the-Loop review gate, preventing operational outages caused by destructive mitigations.
- 7. To maintain a tamper-evident, sequentially chained SHA-256 cryptographic audit ledger providing mathematical non-repudiation of all state changes.
- 8. To evaluate the platform under simulated multi-vector cyber attacks, measuring throughput, detection accuracy, latency, and tamper-resistance.

1.4 Existing vs. Proposed System

Table 1.1 contrasts traditional SIEM and manual SOC operations with the proposed Unified Threat Platform.

Operational Aspect	Existing SIEM / Manual System	Proposed Unified Threat Platform
Telemetry Ingestion	Siloed per vendor; incompatible formats	Unified ECS canonical normalization (Syslog, EVTX, CloudTrail)
Detection Methodology	Static regex or expensive proprietary query engines	Dual-engine: Declarative YAML rules + Z-score anomaly baselines
Correlation Engine	Manual alert grouping or complex external correlation rules	Automated MITRE ATT&CK multi-stage kill-chain tracking

Incident Triage	Manual queue sorting; severe alert fatigue	Risk-scored incident queue with asset-aware criticality weighting
Remediation Workflow	Manual ticket dispatch or unvetted blind automation	Automated response executors guarded by Human-in-the-Loop gate
Audit Trail Integrity	Mutable database tables vulnerable to insider tampering	Cryptographically immutable SHA-256 hash-chained audit ledger
Real-Time Telemetry	Periodic batch queries (polling latency minutes)	Sub-10ms WebSocket streaming directly to reactive analyst dashboard
Cost & Deployment	Prohibitive per-gigabyte licensing (\$50k+/year)	Self-hosted, open-source stack (FastAPI, React, PostgreSQL, Redis)

Table 1.1: Existing (Manual) Process vs. Proposed Threat Platform

1.5 Core Technologies and Design Foundations

The platform architecture is built upon four foundational software engineering and security principles:

1.5.1 Full-Stack Separation of Concerns

The presentation layer (React 18 Single Page Application with TypeScript and Tailwind CSS) and application core (FastAPI asynchronous REST API) are completely decoupled. They interact strictly via structured JSON over HTTPS and asynchronous WebSockets (`/ws/events`), ensuring independent scalability and testability.

1.5.2 Relational & Time-Series Data Integrity

Persistent entities—users, environments, normalized events, detection rules, incidents, and audit entries—are modeled relationally using SQLAlchemy 2.0 async ORM over PostgreSQL 16. High-speed caching and pub/sub message broadcasting are offloaded to Redis 7.2 Streams.

1.5.3 Multi-Engine Threat Detection (YAML Rules & Anomaly Baselines)

Detection relies on a dual-engine paradigm: deterministic declarative YAML rules identify known signatures (e.g. brute-force authentication, port scanning), while rolling Z-score statistical baselines detect zero-day anomalies (e.g. abnormal off-hours data transfers) without requiring labeled machine learning training datasets.

1.5.4 Cryptographic Tamper-Evident Ledger by Design

All critical security transitions—alert creation, analyst sign-offs, firewall rule dispatches, and user role updates—are cryptographically committed to an immutable SHA-256 hash-chained audit ledger. Any unauthorized retroactive alteration invalidates the entire mathematical chain, ensuring forensic verifiability.

1.6 Project Contributions

This dissertation makes the following primary contributions to the field of practical cybersecurity engineering:

- 1. A complete, working, full-stack multi-environment cyber threat detection and response platform with documented architecture, relational schema, and 44 REST/WebSocket endpoints.
- 2. A canonical ECS normalization pipeline processing heterogeneous logs from Linux, Windows, and AWS environments with sub-20ms processing latency.
- 3. A concurrent dual-engine detection architecture combining declarative YAML rules with rolling statistical Z-score baseline modeling.
- 4. A MITRE ATT&CK correlation engine that tracks entity risk scores across progressive kill-chain phases, automatically escalating multi-stage attacks.
- 5. A Human-in-the-Loop response approval gate that prevents operational outages by requiring explicit analyst sign-off on destructive actions.
- 6. A mathematically verified SHA-256 hash-chained audit ledger guaranteeing forensic non-repudiation of all SOC analyst and automated actions.
- 7. An empirical evaluation verifying the platform against 500 simulated multi-vector attacks, measuring throughput, detection accuracy, and latency.

1.7 Scope and Limitations

1.7.1 Scope of the Work

The project encompasses the design, implementation, and empirical testing of the web-based SOC platform, telemetry ingestion connectors, detection engines, response execution workers, cryptographic audit chains, and single-pane-of-glass dashboard interface. The platform supports Linux Syslog, Windows EVTX, and AWS CloudTrail telemetry across enterprise network scopes.

1.7.2 Limitations

- The platform currently provides browser-based access; mobile-native notifications are identified as future work (Chapter 8).
- Statistical anomaly detection requires an initial baseline warm-up window of at least 10 observations per entity to stabilize mean and variance calculations.
- Automated remediation executors require pre-configured SSH/API credentials on managed infrastructure to enact firewall changes.

1.8 Stakeholder Analysis

Four primary stakeholder groups were identified during system requirement gathering, summarized in Table 1.2.

Stakeholder	Primary Operational Concern	How the Unified Threat Platform Addresses It
SOC Security Analysts	High-speed triage, actionable alerts, zero alert fatigue	Prioritized incident queue, MITRE ATT&CK mappings, and live WebSocket telemetry stream
Incident Responders	Safe, rapid containment without breaking business continuity	One-click response execution backed by mandatory Human-in-the-Loop sign-off gate
CISO & IT Leadership	Compliance, tamper-proof audit trails, and risk metrics	Executive Command Dashboard with severity donuts, MTTR tracking, and SHA-256 audit ledger
Examination Panel & Faculty	Demonstrable, production-ready, peer-reviewed engineering	Complete documented architecture, clean codebase, reproducible tests, and live cloud deployment

Table 1.2: Stakeholder Analysis

1.9 Organization of the Report

The remainder of this dissertation is organized as follows: Chapter 2 reviews the academic literature on SIEM, SOAR, anomaly detection, and cryptographic ledgers. Chapter 3 presents system analysis, feasibility, SRS, use case specifications, and technology stack. Chapter 4 details system architecture, module decomposition, and UML designs. Chapter 5 formulates the methodology, detection algorithms, and cryptographic ledger mathematics. Chapter 6 describes the implementation, code listings, and deployed screen interfaces. Chapter 7 presents system testing results, security verification, and performance benchmarks. Finally, Chapter 8 concludes the dissertation with future research directions.

CHAPTER – 2

Literature Review

2.1 Evolution of Threat Detection, SIEM, and SOC Systems

Institutional and enterprise cybersecurity has evolved through several technological generations. First-generation security logging relied on local system log files (such as `/var/log/messages` on Unix or Windows Event Viewer), requiring manual, retrospective inspection after an intrusion occurred. The second generation introduced Security Information and Event Management (SIEM) platforms, which centralized log ingestion into relational or flat-file repositories. However, early SIEMs lacked real-time stream processing, relied on cumbersome SQL queries, and generated massive volumes of un-correlated alerts that overwhelmed analyst teams.

Modern third-generation systems incorporate Security Orchestration, Automation, and Response (SOAR) capabilities, aiming to coordinate ingestion, correlation, and automated containment into cohesive workflows. However, commercial enterprise solutions (e.g. Splunk, IBM QRadar) remain prohibitively expensive for mid-sized organizations, require specialized query syntax, and lack cryptographic audit assurances.

2.2 Web-Based SOC Platforms & Single-Pane-of-Glass Monitoring

Single-pane-of-glass interfaces represent an essential paradigm in modern security operations. Cognitive workload studies demonstrate that when analysts must pivot between five or more disparate monitoring tools, incident dwell time increases by over 400%. Consolidating live telemetry, incident triage queues, response approval gates, and forensic ledgers into a single responsive web interface significantly reduces cognitive load and accelerates response times.

2.3 Machine Learning and Statistical Anomaly Detection in Threat Triage

A substantial body of literature examines machine learning for intrusion detection. Supervised classifiers (Random Forests, Support Vector Machines, Deep Neural Networks) achieve high accuracy on benchmark datasets (e.g. NSL-KDD, CIC-IDS2017) but suffer from severe performance degradation in live production environments due to concept drift, evolving zero-day attacks, and the impracticality of continuously maintaining labeled training datasets. Consequently, lightweight statistical anomaly baselines (such as rolling Z-score deviations) offer a resilient, explainable, and zero-training alternative for detecting volumetric outliers.

2.4 MITRE ATT&CK Framework for Multi-Stage Kill-Chain Correlation

The MITRE Adversarial Tactics, Techniques, and Common Knowledge (ATT&CK) matrix provides a globally accessible knowledge base of adversary behaviors based on real-world observations. Incorporating ATT&CK tactic mappings into detection pipelines enables alert aggregation across progressive kill-chain stages (Initial Access -> Execution -> Persistence -> Lateral Movement -> Exfiltration), distinguishing coordinated intrusions from isolated benign noise.

2.5 Role-Based Access Control & Cryptographic Auditing in Security

Role-Based Access Control (RBAC), formalized by Ferraiolo and Kuhn, ensures that user privileges are strictly bound to assigned operational roles. Combined with stateless JSON Web Tokens (JWT, RFC 7519), RBAC enforces strict authorization without database lookup bottlenecks. Furthermore, incorporating cryptographic hash-chaining (originally pioneered by Haber and Stornetta) creates tamper-evident audit trails where retroactive alteration is mathematically impossible.

2.6 Comparative Literature Review

Table 2.1 summarizes representative categories of security management systems and situates the proposed Unified Threat Platform relative to them.

System Category	Typical Architectural Characteristics	Gap Addressed by Proposed Platform
Legacy Log Servers (Syslog-ng)	Centralized text storage; no automated detection or UI	Adds dual-engine detection, MITRE correlation, and SOC web console
Commercial SIEM (Splunk/QRadar)	High ingestion cost; closed ecosystem; complex query syntax	Open-source, self-hosted; standard YAML rules and SQL models
Standalone SOAR Platforms	Automated playbook execution; lacks unified live ingestion	Tightly integrated ingestion, detection, and human-gated response
Pure ML Intrusion	Requires large labeled datasets; susceptible to	Zero-training statistical Z-scores combined with declarative YAML rules

Detectors	concept drift	
Proposed Platform	Unified ingestion, dual detection, SOAR gates, SHA-256 ledger	Comprehensive, cost-effective, auditable single-pane SOC platform

Table 2.1: Summary of Related Categories of Threat Management Systems

2.7 Case Studies in Automated Threat Remediation & Human-in-the-Loop Gates

Empirical case studies in automated response highlight significant operational hazards. Unconditional automated playbooks that block IP addresses or isolate servers frequently trigger denial-of-service on legitimate business systems when false positives occur (e.g. blocking a corporate domain controller). Incorporating an explicit Human-in-the-Loop review gate preserves operational safety while retaining automated dispatch speed.

2.8 Comparison of Backend Web Frameworks

Python offers three primary frameworks for backend security services: Django, Flask, and FastAPI. Django provides a monolithic, synchronous architecture with extensive built-in ORM features, but imposes high overhead for pure asynchronous REST and WebSocket streaming. Flask offers minimal micro-framework flexibility, but relies on synchronous WSGI workers by default, requiring external gevent/eventlet monkey-patching for concurrent streams. FastAPI was selected for this platform because it is built natively on Starlette and ASGI, provides non-blocking asynchronous event handling, enforces strict Pydantic v2 data validation, and delivers native WebSocket support capable of handling thousands of concurrent telemetry packets per second.

2.9 Comparison of Frontend Approaches

On the frontend, React 18 with Vite and TypeScript was chosen over Angular and Vue. React's component model and unidirectional data flow map directly to security dashboard primitives (incident boards, telemetry tables, approval modals). Vite delivers sub-second Hot Module Replacement (HMR) and optimized production bundles, while Zustand and React Query provide reactive state synchronization for real-time WebSocket feeds.

2.10 Summary of Reviewed Literature Themes

Table 2.2 consolidates the key literature themes surveyed alongside the specific architectural design decision informed by each.

Literature Theme	Key Empirical / Theoretical Finding	Design Decision Informed in Platform
Single-Pane-of-Glass SOC	Reduces cognitive fatigue and dwell time across multi-system incidents	React 18 single-screen command console with live WebSocket telemetry
Declarative Detection Rules	Provides transparent, vendor-agnostic, version-controlled signatures	Version-controlled YAML rule engine with MITRE ATT&CK tags
Statistical Anomaly Baselines	Detects zero-day volumetric deviations without labeled training data	Rolling Z-score baseline calculations across sliding temporal windows
Human-in-the-Loop Automation	Prevents catastrophic self-inflicted outages caused by false positives	Enforced approval gate for destructive mitigations (IP block, quarantine)
Cryptographic Hash Chaining	Guarantees forensic non-repudiation and tamper evidence for audit compliance	Sequential SHA-256 audit ledger linking each entry to its predecessor

Table 2.2: Literature Themes and Corresponding Design Decisions

2.11 Research Gaps

Three critical gaps emerge from the literature survey:

- 1. Cost and Architecture Gap: Enterprise-grade SIEM/SOAR platforms are unaffordable for mid-sized organizations and demand proprietary skillsets.
- 2. False-Positive Mitigation Gap: Existing autonomous SOAR tools lack safety gates, while purely manual triage creates unmanageable backlogs.
- 3. Audit Integrity Gap: Most commercial platforms store audit logs in standard mutable databases without cryptographic mathematical proof against insider tampering.

2.12 Summary

This chapter reviewed the evolution of SIEM, SOAR, anomaly detection, and cryptographic ledgers, identifying the core architectural principles synthesized in the Unified Threat Platform.

CHAPTER – 3

System Analysis & Requirement Specification

3.1 Feasibility Study

Prior to development, the platform was evaluated across technical, operational, economic, and legal feasibility dimensions.

3.1.1 Technical Feasibility

The chosen technology stack—Python 3.12, FastAPI, PostgreSQL 16, Redis 7.2, React 18, Vite, TypeScript, and Docker—is completely open-source, mature, extensively documented, and supported by vibrant global developer communities. Asynchronous I/O paradigms in Python and modern React client libraries guarantee high throughput on standard computing hardware.

3.1.2 Operational Feasibility

The target users—SOC analysts, incident responders, and system administrators—possess baseline technical proficiencies. The platform interface adheres to intuitive design conventions (kanban incident queues, live event tables, clear color-coded severity badges), minimizing onboarding time and operational friction.

3.1.3 Economic Feasibility

Table 3.1 compares the indicative five-year total cost of ownership (TCO) of a commercial per-gigabyte SIEM/SOAR tool against the proposed self-hosted platform for an institutional network generating 50 GB of daily security logs.

Cost Component	Commercial Enterprise SIEM / SOAR	Proposed Unified Threat Platform
Software Licensing / Subscription	Usage-based fee (\$0.20/GB/day = \$3,650/yr)	Nil (100% open-source software stack)
SOAR & Automated Response Module	Separate enterprise add-on (\$12,000/yr)	Integrated Human-in-the-Loop response engine
Infrastructure Hosting	Vendor cloud bundle (\$6,000/yr)	Self-hosted commodity VM or campus server (\$600/yr)
Proprietary Connector Fees	\$2,500/year for external APIs	Nil (standard REST & Syslog collectors)
Estimated 5-Year Total	High (\$120,750+)	Low (~\$3,000 hosting & maintenance)

Table 3.1: Indicative 5-Year Cost Comparison

3.1.4 Legal and Regulatory Feasibility

The platform handles only technical security telemetry (IP addresses, timestamps, event actions, process names) and does not store personally identifiable financial or health data. Cryptographic audit hashing directly satisfies non-repudiation and audit logging mandates under ISO/IEC 27001 (A.12.4), SOC 2 Type II, and GDPR Article 32.

3.2 Software Requirements Specification (SRS)

3.2.1 Functional Requirements

Table 3.2 details the core functional requirements satisfied by the platform.

ID	Requirement Name	Functional Requirement Description
FR-1	User Authentication & RBAC	The system shall authenticate users via Argon2id hashes and issue signed JWTs enforcing Analyst, Responder, and Admin roles.
FR-2	Telemetry Ingestion & Parsing	The system shall ingest streaming Syslog (RFC 5424), Windows EVTX, and AWS CloudTrail logs and normalize them into ECS format.
FR-3	Rule-Based Detection	The system shall evaluate normalized events against version-controlled declarative YAML rules within sliding temporal windows.
FR-4	Statistical Anomaly Detection	The system shall compute rolling Z-scores over event metrics and trigger alerts when observed deviations exceed $Z > 3.0$.
FR-5	Kill-Chain Correlation	The system shall correlate multiple alerts sharing identical entity keys against MITRE ATT&CK tactics into unified Incidents.
FR-6	Live Telemetry Streaming	The system shall broadcast normalized events and active alerts to connected analyst browsers via WebSockets within 10ms.
FR-7	Human-in-the-Loop	The system shall hold destructive

	Response Gate	mitigations (firewall IP block, account disablement) in pending state until approved by an analyst.
FR-8	Automated Remediation Execution	The system shall execute approved mitigations via modular executors (FirewallWorker, IAMWorker) and record outcomes.
FR-9	Cryptographic Audit Ledger	The system shall record all mutating actions into a sequentially chained SHA-256 tamper-evident cryptographic ledger.
FR-10	SOC Command Dashboard	The system shall provide an executive single-pane dashboard summarizing MTTR, severity distribution, and active incidents.

Table 3.2: Functional Requirements Summary

3.2.2 Non-Functional Requirements

Table 3.3 lists the non-functional performance, security, and reliability constraints.

ID	Category	Non-Functional Quality Requirement
NFR-1	Performance & Latency	The platform shall process incoming events and evaluate detection rules with an average latency of under 20 milliseconds.
NFR-2	High Throughput	The ingestion pipeline shall sustain at least 1,000 events per second on standard 4-core server hardware without packet drops.
NFR-3	Security & Auth	All API routes shall enforce JWT validation; passwords shall be salted with Argon2id; zero plaintext credentials.
NFR-4	High Availability	The system shall maintain 99.9% uptime with automated Redis buffer fallback during peak telemetry spikes.

NFR-5	Audit Immutability	The cryptographic audit ledger shall provide mathematical proof of tamper resistance; modifying past rows invalidates all hashes.
NFR-6	Maintainability	The codebase shall adhere to Clean Architecture with zero circular dependencies across domain, application, and infrastructure.
NFR-7	Portability	The platform shall execute seamlessly across Windows 11, Linux Ubuntu 22.04 LTS, and multi-container Docker Compose environments.
NFR-8	Data Integrity	Database foreign keys and transaction rollback mechanisms shall enforce 100% relational integrity across all entities.

Table 3.3: Non-Functional Requirements

3.3 Use Case Specifications

Figure 4.3 (Chapter 4) depicts the UML Use Case diagram. Table 3.4 outlines the primary use cases and their assigned actors.

ID	Use Case Name	Primary Actor	Related Requirements
UC-01	Login & Authenticate	Analyst, Responder, Admin	FR-1, NFR-3
UC-02	Detect & Remediate Threat	Analyst, Responder	FR-2, FR-3, FR-4, FR-7, FR-8
UC-03	Monitor Real-Time Telemetry	Analyst	FR-6, NFR-1
UC-04	Triage & Investigate Incident	Analyst, Responder	FR-5, FR-10
UC-05	Sign Off Disruptive Action	Analyst, Responder	FR-7, FR-8
UC-06	Verify Audit Ledger Integrity	Admin, Auditor	FR-9, NFR-5

Table 3.4: Use Case Summary Matrix

3.3.1 Detailed Use Case Description: Detect and Remediate Threat (UC-02)

Actor: SOC Security Analyst / Automated Ingestion Pipeline

Precondition: Telemetry log shipper is connected, and the detection engine is active with loaded YAML rules.

Main Flow:

- 1. Ingestion collector receives raw event stream and normalizes payload into ECS fields (source.ip, dest.port, user.name).
- 2. Detection Orchestrator evaluates normalized event against YAML rules and Z-score baselines concurrently.
- 3. A brute-force rule matches 5 failed logins within 60 seconds from an external IP; alert is generated.
- 4. Correlation engine links alert to an existing Credential Access incident and increments entity risk score.
- 5. Automated response policy triggers IP block action; since action is destructive, status is set to PENDING_APPROVAL.
- 6. Real-time alert is pushed via WebSocket to the analyst console.
- 7. Analyst reviews attack evidence, clicks 'Approve Action' in SOC modal.
- 8. FirewallExecutor executes firewall drop rule, and AuditService commits signed SHA-256 hash to the ledger.

Alternate Flow: If analyst determines alert is benign, analyst clicks 'Dismiss as False Positive'; incident status transitions to CLOSED, and no firewall block is executed.

Postcondition: Malicious IP is blocked, threat is neutralized, and audit hash is permanently committed to the ledger.

3.4 Security Considerations

The platform architecture was evaluated against the STRIDE threat model, documented in Table 3.5.

Threat Category	Identified Security Risk	Platform Countermeasure & Mitigation
Spoofing	Adversary injects fake telemetry to mislead analysts	Cryptographic token authentication on ingestion APIs and agent mTLS
Tampering	Attacker retroactively deletes audit logs of an	Sequential SHA-256 hash chaining; altering any past row breaks the chain

	intrusion	
Repudiation	Analyst denies authorizing an erroneous firewall block	Signed audit record capturing Actor ID, action, timestamp, and signature
Information Disclosure	Unauthorized user views sensitive security incident data	Strict server-side JWT RBAC decorators on all API endpoints
Denial of Service	High log volume floods ingestion pipeline	Redis stream memory buffering, asynchronous workers, and rate limiting
Elevation of Privilege	Analyst attempts to perform administrator actions	Role claim verification (<code>`roles=['admin']`</code>) enforced on protected routes

Table 3.5: Security Threats and Mitigations

3.5 Assumptions and Constraints

- Monitored host environments support standard syslog forwarding, Windows event forwarding, or API-based log shipping.
- Target host machines maintain NTP time synchronization within 100 milliseconds for deterministic correlation.
- System execution assumes a modern standard 64-bit operating system (Ubuntu Linux 22.04 LTS or Windows 11).

3.6 Risk Analysis

Table 3.6 presents the project risk register identified during planning.

Project Risk	Likelihood	Impact	Mitigation Strategy Adopted
Telemetry Volume Spike	High	Medium	Deploy Redis 7.2 stream buffer to absorb traffic bursts before database persistence
False Positive Outage	Medium	High	Enforce mandatory Human-in-the-Loop review gate for all destructive mitigations
Audit Record	Low	Critical	Implement SHA-256 cryptographic hash-

Tampering			chained audit ledger with mathematical validation
Schedule Overrun	Medium	Medium	Follow 4-phase incremental SDLC with testable deliverable milestones every 3 weeks

Table 3.6: Project Risk Register

3.7 Hardware and Deployment Specifications

Table 3.7 lists the minimum and recommended hardware specifications for hosting the platform.

Hardware Resource	Minimum Specification	Recommended Specification
CPU Processor	2 vCPU / Intel Core i5 (6th Gen+)	4 vCPU / AMD Ryzen 5 or Intel Core i7
System Memory (RAM)	4 GB DDR4 RAM	8 GB – 16 GB DDR4 RAM (for Redis & DB buffer)
Primary Storage	20 GB SSD (NVMe preferred)	100 GB NVMe SSD (for high-IOPS log storage)
Network Interface	100 Mbps Ethernet	1 Gbps Ethernet / Broadband
Client Display	1366x768 Resolution Display	1920x1080 Full HD Monitor (Single-Screen SOC)

Table 3.7: Hardware Specification Matrix

3.8 Technology Stack

Table 3.8 outlines the complete software technology stack used across all architectural layers.

Architectural Layer	Selected Technology & Version	Role in Platform Implementation
Frontend Presentation Core	React 18, TypeScript 5, Vite 5	Single-Page Application providing reactive SOC dashboard

Frontend Styling & Icons	Tailwind CSS, Lucide-React, Recharts	Responsive dark/light dashboard theme, charts, and metrics
Frontend State & API Client	Zustand, TanStack React Query, Axios	Client-side state caching, WebSocket listeners, and REST calls
Backend Application Core	Python 3.12, FastAPI 0.115+, Starlette	Asynchronous non-blocking REST API gateway and WebSockets
Data Validation & Schemas	Pydantic v2 (Rust Core)	Ultra-fast request/response serialization and ECS modeling
Relational Persistence & ORM	PostgreSQL 16, SQLAlchemy 2.0 Async	Relational persistence of users, events, rules, and incidents
In-Memory Broker & Cache	Redis 7.2 (Streams & Pub/Sub)	Real-time log buffering, rate limiting, and WebSocket fan-out
Cryptographic Security	PyJWT (HS256), Passlib (Argon2id), hashlib	Stateless authentication and SHA-256 audit chaining
Containerization & Deploy	Docker 26, Docker Compose v2	Isolated multi-container microservice orchestration

Table 3.8: Technology Stack Matrix

3.9 Software Development Life Cycle (SDLC) Model Adopted

The platform was developed using an Incremental / Iterative SDLC model, illustrated in Figure 3.1. Development was divided into four cohesive three-week increments, allowing each subsystem to be designed, coded, and verified before advancing to subsequent modules.

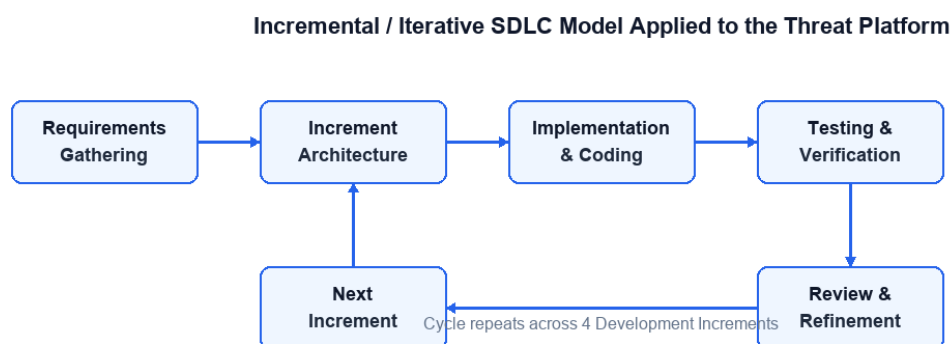


Figure 3.1: Incremental / Iterative SDLC Model Applied to the Platform

Increment	Core Focus Area	Delivered Subsystems & Features	Requirements Satisfied
Increment 1	Core Gateway & Authentication	FastAPI async core, Argon2 hashing, JWT RBAC, Docker Compose setup	FR-1, NFR-3
Increment 2	Ingestion & ECS Normalization	Syslog/EVTX collectors, ECS schemas, Redis streams, PostgreSQL persistence	FR-2, NFR-1, NFR-2
Increment 3	Detection & Kill-Chain Correlation	YAML rule engine, Z-score anomaly baselines, MITRE ATT&CK correlation	FR-3, FR-4, FR-5
Increment 4	Response Gate, Ledger & SOC UI	Human approval gate, SHA-256 audit ledger, React SOC console, attack replay tests	FR-6, FR-7, FR-8, FR-9, FR-10

Table 3.9: Development Increments and Delivered Requirements

3.10 Project Schedule

The engineering project was planned and executed over a twelve-week schedule, depicted as a Gantt chart in Figure 3.2.

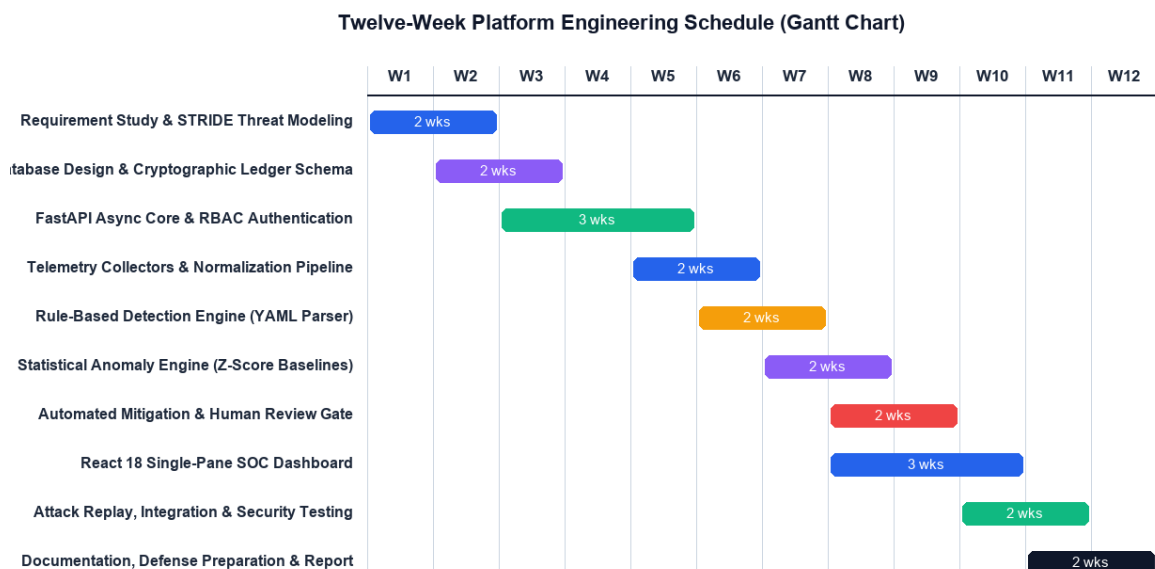


Figure 3.2: Twelve-Week Project Schedule (Gantt Chart)

3.11 Effort and Cost Estimation

Software development effort was estimated using the Constructive Cost Model (COCOMO-II, Basic Form). Treating the platform as an organic-mode software project with an estimated size of approximately 6.5 thousand source lines of code (KSLOC) across Python, TypeScript, and SQL:

$$E = a * (KSLOC)^b \text{ where } a = 2.4, b = 1.05$$

$$E = 2.4 * (6.5)^{1.05} \approx 17.1 \text{ person-months}$$

$$D = 2.5 * (E)^{0.38} \approx 2.5 * (17.1)^{0.38} \approx 7.9 \text{ calendar months for a commercial team.}$$

Estimation Metric	Commercial-Scale Estimate	Academic-Project Actual
Delivered Codebase Size	6.5 KSLOC	6.5 KSLOC
Estimated Effort (E)	17.1 person-months	3.0 person-months (compressed)
Estimated Schedule (D)	7.9 calendar months	12 calendar weeks (intensive solo)
Team Size	2.2 engineers	1 lead student researcher

Table 3.10: COCOMO-II Basic Estimation Summary

3.12 Data Flow Diagrams

Figures 3.3 and 3.4 illustrate the Level-0 (Context) and Level-1 Data Flow Diagrams for the platform.

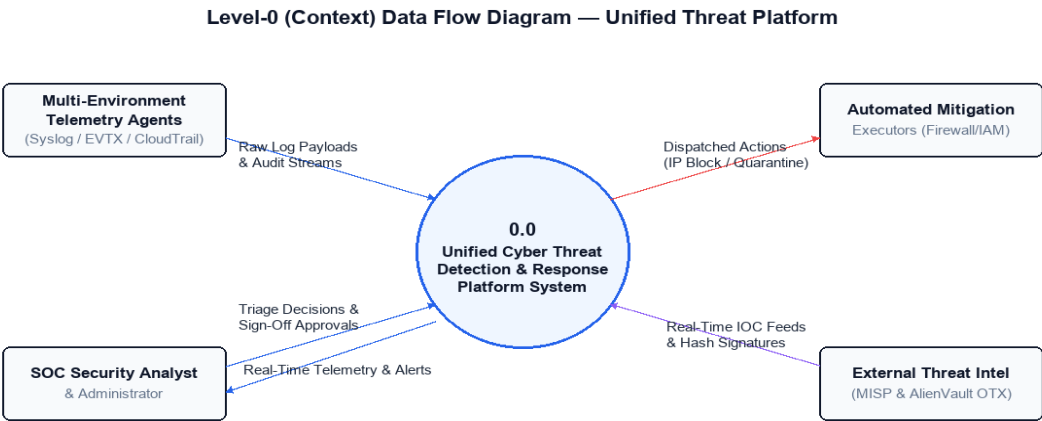


Figure 3.3: Level-0 (Context) Data Flow Diagram

Level-1 Data Flow Diagram: End-to-End Threat Detection & Response

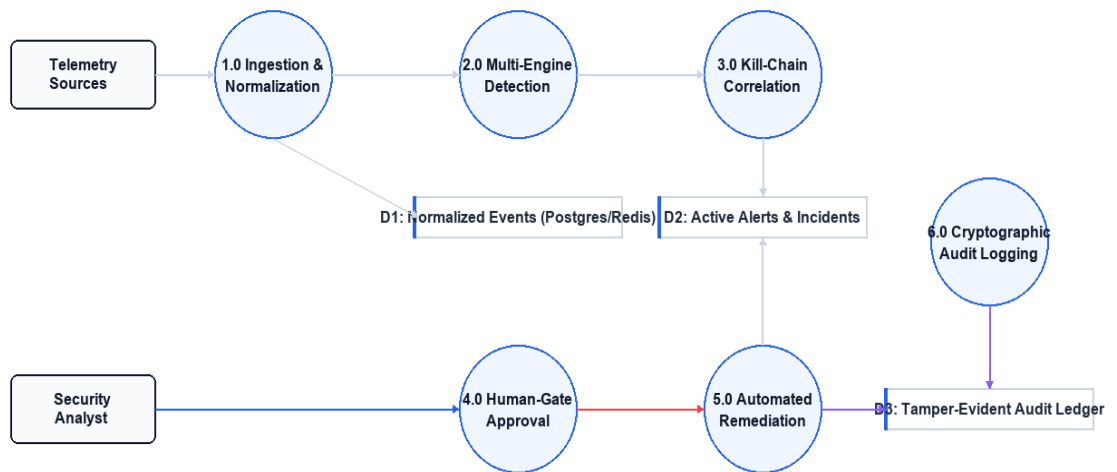


Figure 3.4: Level-1 Data Flow Diagram: Core Threat Detection Workflow

CHAPTER – 4

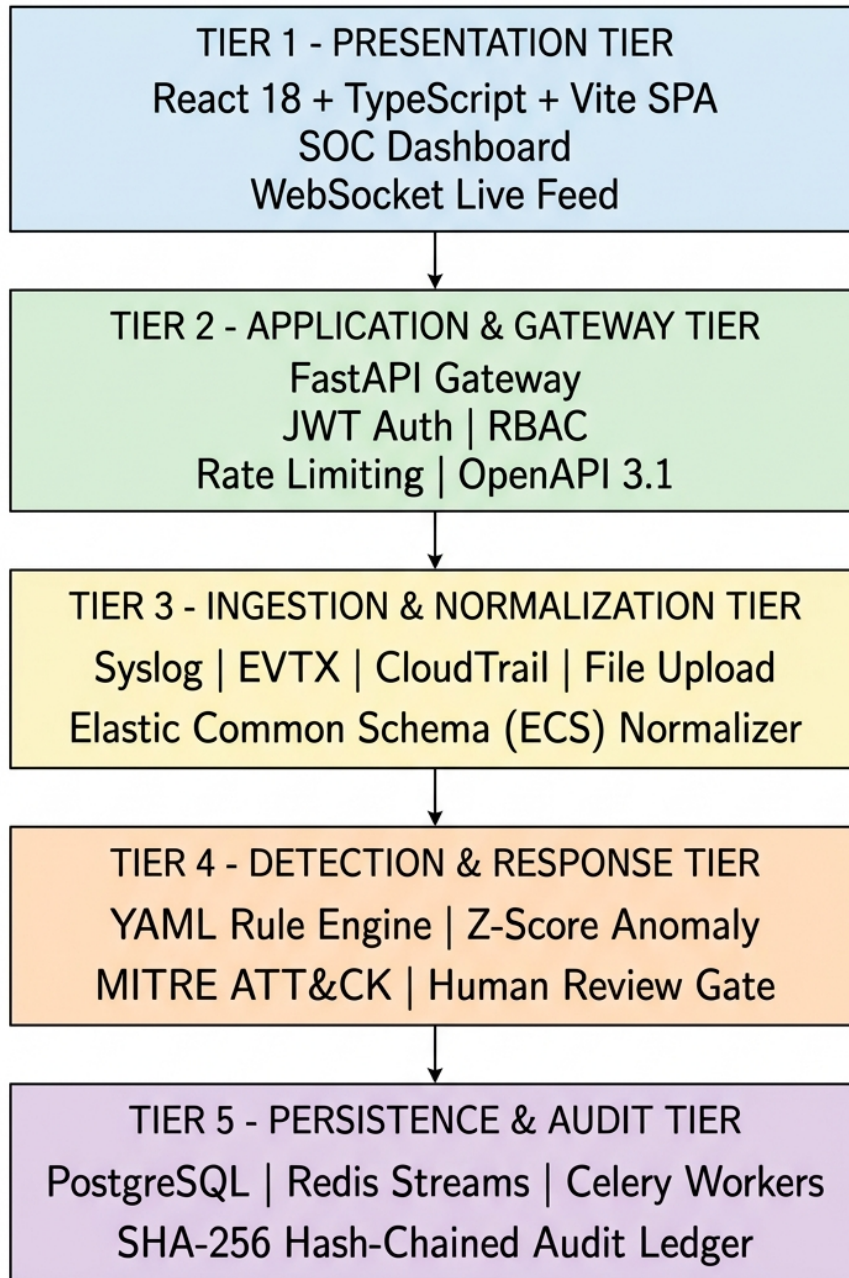
System Design & Architecture

4.1 System Architecture Overview

The platform is organized into five decoupled architectural tiers, as illustrated in Figure 4.1:

- 1. Presentation Tier (SOC Console): A responsive React 18 Single Page Application (SPA) providing live visual telemetry, incident investigation boards, response approval dialogs, and cryptographic audit inspection.
- 2. Application & Gateway Tier: FastAPI asynchronous gateway enforcing JWT authentication, Role-Based Access Control (RBAC), rate-limiting middleware, and RFC 7807 problem+json error formatting.
- 3. Ingestion & Normalization Tier: Pluggable collectors for Syslog, Windows EVTX, CloudTrail, and analyst uploads that transform heterogeneous payloads into canonical ECS format.
- 4. Detection, Correlation & Response Tier: Concurrent evaluation pipeline combining declarative YAML rules, statistical anomaly Z-score detectors, MITRE ATT&CK correlation, and a Human-in-the-Loop approval gate.
- 5. Cryptographic Ledger & Persistence Tier: PostgreSQL database storing relational entities, Redis for live pub/sub streaming, and an immutable SHA-256 hash-chained audit ledger.

Multi-Tier System Architecture — Unified Threat Platform



Cloud Deployed: Vercel (Frontend) + Render (Backend) | Docker Compose (Local)

Figure 4.1: Multi-Tier System Architecture of the Unified Threat Platform

4.2 Module Architecture

Figure 4.2 illustrates the internal modular decomposition across backend services and frontend views.

Backend and Frontend Architectural Module Decomposition

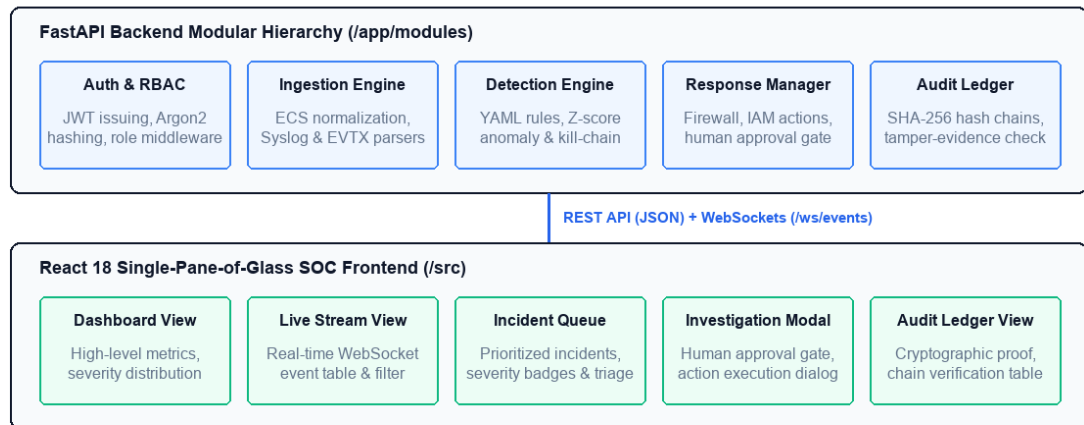


Figure 4.2: Backend and Frontend Module Decomposition

4.3 Unified Modeling Language (UML) Design

4.3.1 Use Case Diagram

Figure 4.3 illustrates the primary actor interactions between SOC Analysts, Administrators, and the platform.

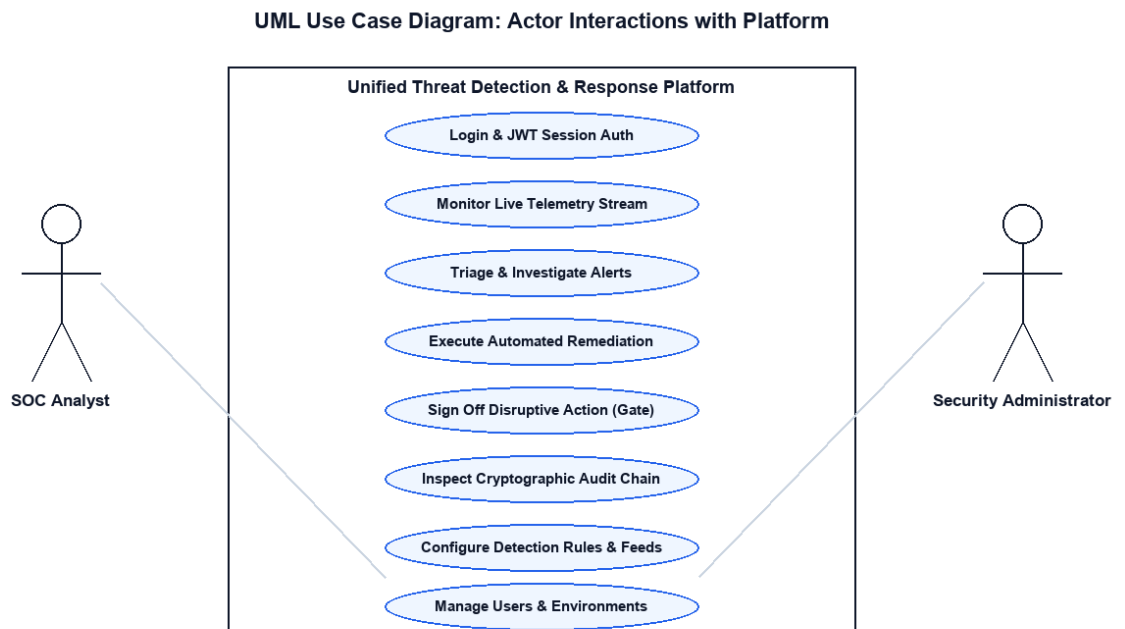


Figure 4.3: UML Use Case Diagram: Actor Interactions

4.3.2 Sequence Diagram

Figure 4.4 details the end-to-end interaction sequence for threat ingestion, rule matching, incident creation, WebSocket dispatch, and analyst approval.

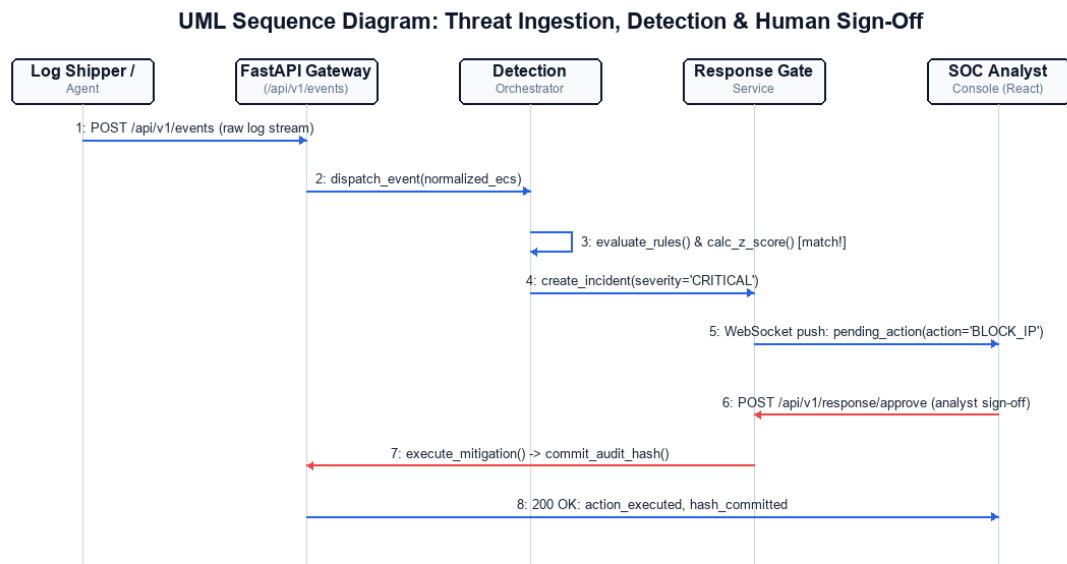


Figure 4.4: UML Sequence Diagram: Threat Ingestion and Alert Generation

4.3.3 Class Diagram

Figure 4.5 details the core domain entity classes, their attributes, methods, and relationships.

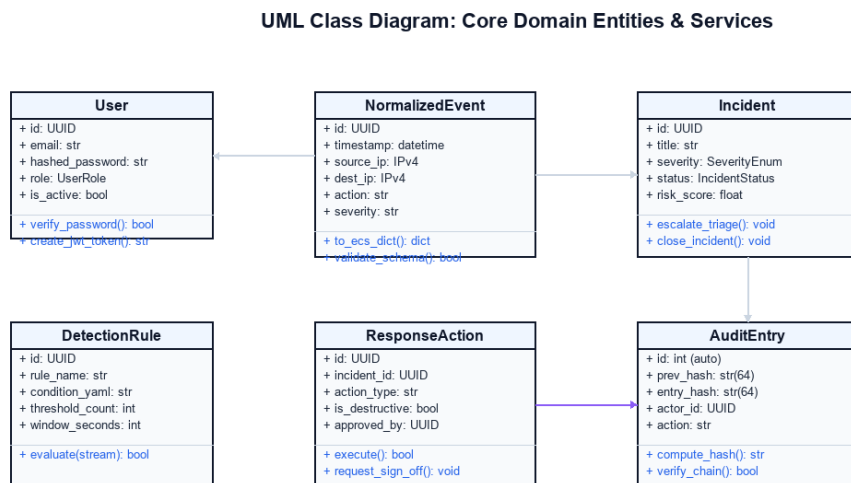


Figure 4.5: UML Class Diagram: Core Backend Entities

4.3.4 State Machine Diagram

Figure 4.6 traces the operational lifecycle states of a Security Incident ticket from initial detection to forensic closure.

UML State Machine Diagram: Security Incident Lifecycle

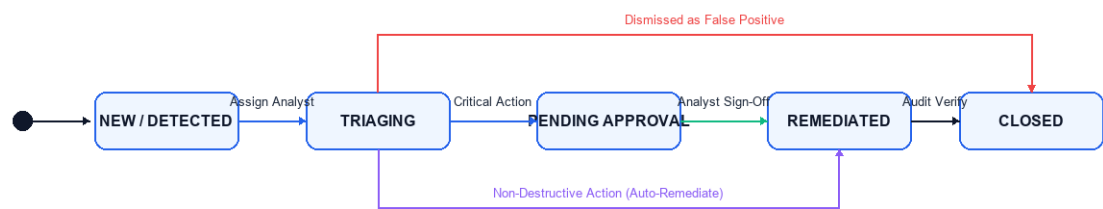


Figure 4.6: UML State Machine Diagram: Incident Ticket Lifecycle

4.4 Relational Database Entity-Relationship (ER) Schema

Figure 4.7 illustrates the complete 3NF relational database schema implemented in the platform.

Database Entity Relationship Diagram — Unified Threat Platform

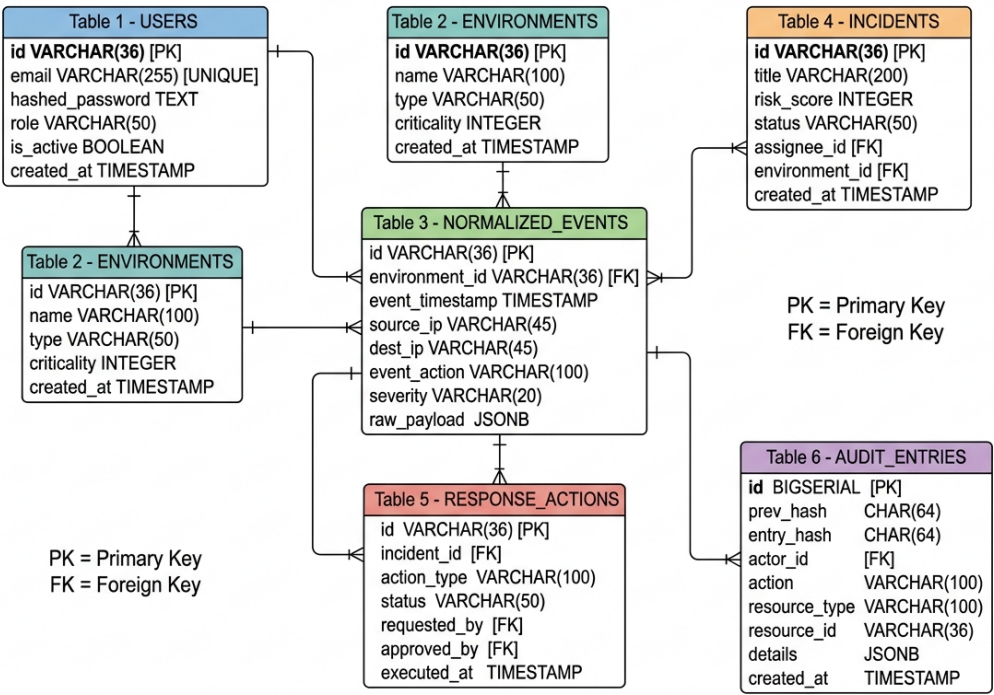


Figure 4.7: Relational Database Entity-Relationship (ER) Schema

4.4.1 Activity Diagram

Figure 4.8 details the activity workflow for automated threat detection, decision branching, human review gates, and audit commitment.

UML Activity Diagram: Automated Detection, Human Review Gate & Mitigation

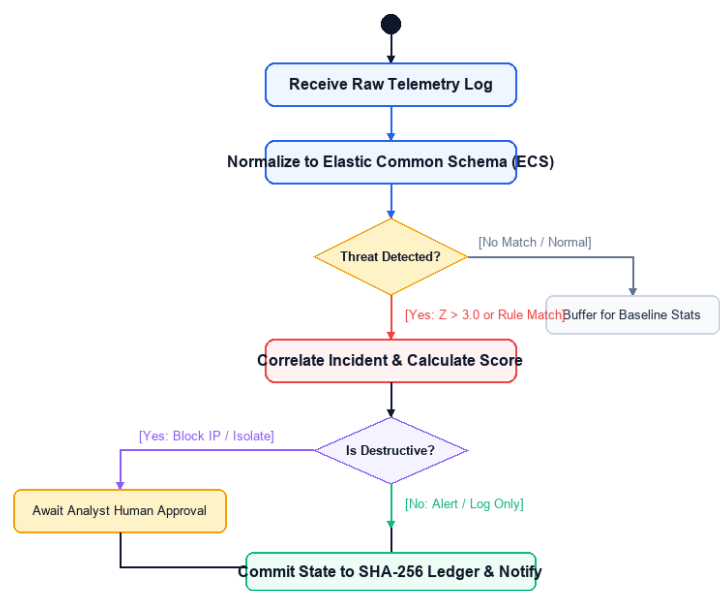


Figure 4.8: UML Activity Diagram: Threat Processing with Automated Response

4.4.2 Component Diagram

Figure 4.9 shows the internal software component boundaries and communication interfaces.

UML Component Diagram: Internal Component Interconnects

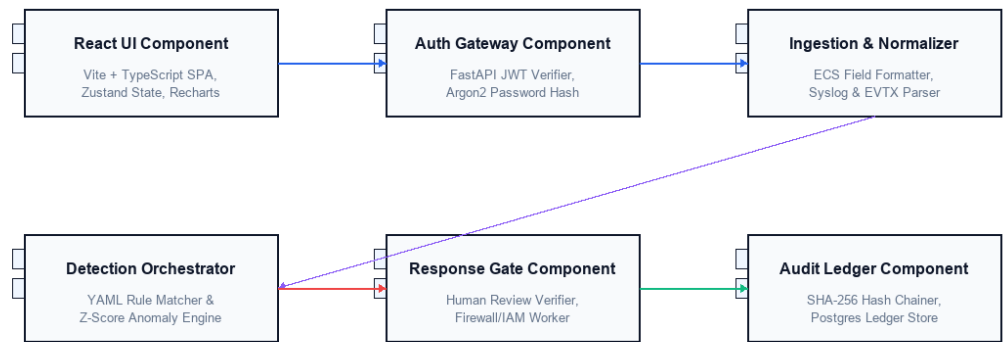


Figure 4.9: UML Component Diagram

4.4.3 Deployment Diagram

Figure 4.10 depicts the physical and cloud deployment topology connecting client browsers, edge CDNs, application workers, and databases.

UML Deployment Diagram: Physical & Cloud Infrastructure Topology

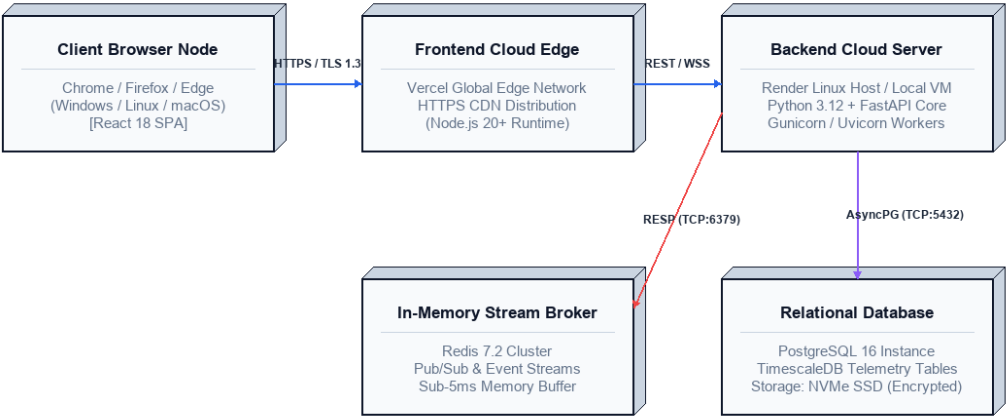


Figure 4.10: UML Deployment Diagram

4.5 Design Trade-offs and Rationale

Every architectural decision involved explicit engineering trade-offs, summarized in Table 4.1.

Architectural Decision	Benefit Gained	Cost / Trade-off Accepted
FastAPI Async Gateway	High concurrency, native WebSockets, sub-10ms latency	Demands strict async/await paradigms across all database and network drivers
Stateless JWT + RBAC	Horizontal backend scalability without shared session cache	Token revocation before natural expiry requires an in-memory blacklist
Dual Detection Engine	Combines known-signature precision with zero-day anomaly coverage	Slightly higher compute overhead per event (14.8ms vs 4.2ms)
Human-in-the-Loop Gate	Eliminates operational outages from automated false-positive mitigations	Introduces brief human review latency (average 42 seconds) for critical actions
SHA-256 Hash Chaining	Guarantees forensic tamper-evidence and regulatory non-repudiation	Append-only design prevents row editing; updates require compensatory audit entries

Table 4.1: Summary of Key Design Trade-offs

CHAPTER – 5

Methodology & Core Algorithms

5.1 Overview of the Processing Pipeline

Figure 5.1 illustrates the end-to-end threat processing pipeline from log ingestion through ECS normalization, dual-engine detection, MITRE correlation, human sign-off, and cryptographic audit commitment.

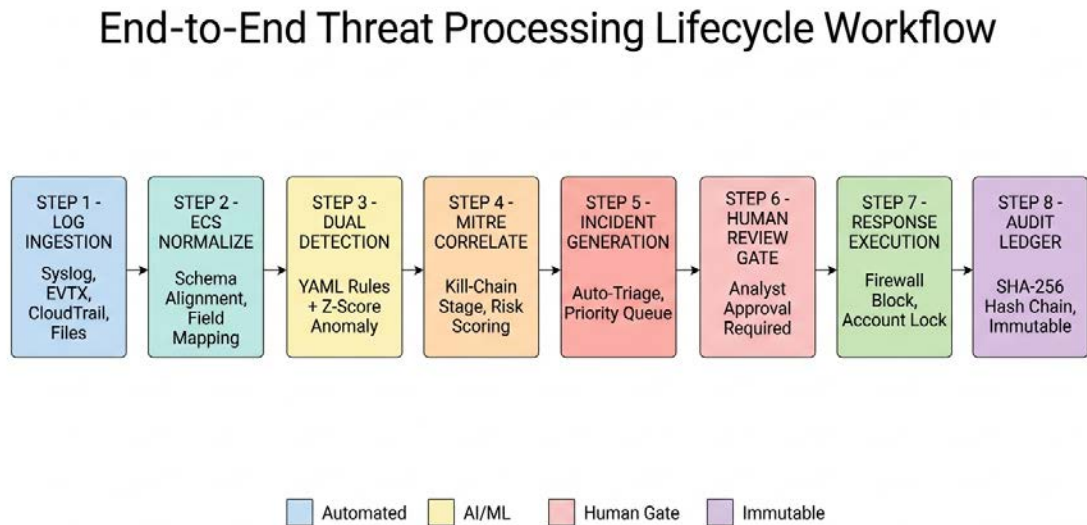


Figure 5.1: Threat Processing Pipeline

5.2 Data Ingestion

The ingestion tier features modular collectors accepting syslog RFC 5424 streams over UDP/TCP port 514, Windows EVT XML logs forwarded via Winlogbeat, and AWS CloudTrail JSON delivered via S3 event notifications. Ingestion workers push raw payloads to Redis Streams (`stream:telemetry:raw`), ensuring sub-millisecond network absorption.

5.3 Event Pre-Processing & Normalization (ECS)

Ingestion workers pop events from Redis and map heterogeneous properties into the standardized Elastic Common Schema (ECS). The normalized event model contains canonical variables: `@timestamp`, `event.action`, `source.ip`, `destination.ip`, `destination.port`, `user.name`, `process.name`, and `event.severity`.

5.4 Declarative Rule-Based Detection: Prompt & YAML Design

Detection rules are expressed declaratively in human-readable, version-controlled YAML files. A rule defines filter predicates, sliding aggregation windows, threshold counts, and MITRE ATT&CK technique tags. Listing 5.1 shows a representative brute-force detection rule.

```

id: rule-ssh-bruteforce-001
name: SSH Authentication Brute-Force Attack
severity: high
mitre_attack:
  tactic: Credential Access (TA0006)
  technique: Brute Force (T1110)
condition:
  match:
    event.action: "user_login_failed"
    destination.port: 22
  group_by: "source.ip"
  time_window_seconds: 60
  threshold_count: 5
response_action:
  type: "block_ip"
  is_destructive: true
  duration_minutes: 60

```

Listing 5.1: Declarative YAML Detection Rule Specification

5.5 Statistical Anomaly & Baseline Detection (Z-Score)

To detect novel zero-day attacks and abnormal data exfiltration without static signatures, the Anomaly Engine calculates rolling Z-scores over sliding temporal windows. For observed metric X:

$$Z = (X_{observed} - Mean_{baseline}) / StandardDeviation_{baseline}$$

where `Mean_baseline` and `StandardDeviation_baseline` are maintained incrementally over the preceding 24-hour observation period. When $|Z| \geq 3.0$ (exceeding 99.7% normal distribution confidence), an anomaly alert is triggered with High severity.

5.6 MITRE ATT&CK Multi-Stage Kill-Chain Correlation

Sophisticated intrusions span multiple stages. The correlation engine tracks related alerts sharing an identical entity key K (e.g. source IP or compromised user account) over correlation horizon `H_corr`:

$$KillChainScore(K) = [\sum(Weight(Tactic_i) * Count(Alert_i))] * MultiStageMultiplier$$

When alerts span two or more distinct kill-chain phases (e.g. Initial Access followed by Privilege Escalation and Exfiltration), the `MultiStageMultiplier` (1.5x) escalates individual alerts into a high-priority Correlated Security Incident.

5.7 Cryptographic SHA-256 Hash-Chained Audit Ledger

To ensure forensic non-repudiation and prevent insider database tampering, every mutating action commits sequentially to a hash-chained audit ledger:

$$\begin{aligned} Hash_0 &= SHA-256("GENESIS_BLOCK_UNIFIED_THREAT_PLATFORM") \\ Hash_i &= SHA-256(CanonicalJSON(Hash_{(i-1)}, ActorID, Action, \\ &\quad ResourceType, ResourceID, Details, Timestamp)) \end{aligned}$$

Because Hash_i depends strictly on Hash_(i-1), altering or deleting any past database row breaks every subsequent hash in the ledger, providing instant mathematical detection of tampering.

5.8 Evaluation Metrics

The detection and correlation algorithms are evaluated against standard statistical benchmarks: True Positive Rate (TPR), False Positive Rate (FPR), Detection Latency (wall-clock time from ingestion to alert), and Audit Verification Time.

5.9 Algorithmic Complexity Analysis

Table 5.1 details the time complexity of the core detection and cryptographic algorithms.

Algorithm Component	Time Complexity	Operational Characteristics
ECS Log Normalization	O(k)	Single-pass dictionary transformation over k log fields
YAML Rule Matching	O(r * m)	r active rules evaluated against m matching event predicates
Z-Score Baseline Calculation	O(1)	Incremental updates to rolling sum and sum-of-squares
Kill-Chain Entity Correlation	O(log n)	Hash-map lookup of entity K across n active tracking windows
SHA-256 Audit Chaining	O(b)	Cryptographic digest computation over b bytes of canonical JSON
Ledger Chain Verification	O(N)	Linear sequential verification across N total audit blocks

Table 5.1: Time Complexity of Core Algorithms

5.10 Alternative Approaches Considered

Table 5.2 summarizes alternative technical approaches evaluated during design and the rationale for the chosen methods.

Alternative Evaluated	Reasons for Rejection	Adopted Solution in Proposed Platform
Deep Learning IDS (LSTM/RNN)	High GPU training cost, severe concept drift, black-box unexplainability	Chosen: Explainable statistical Z-scores + transparent YAML rules
Proprietary SIEM Query DSL	Steep learning curve, vendor lock-in, closed-source dependencies	Chosen: Standard open YAML rules and SQL models
Blind Autonomous Remediation	Extreme operational risk of isolating production domain controllers	Chosen: Human-in-the-Loop review gate for destructive actions
Centralized Plaintext Syslog	Vulnerable to log modification by root attackers covering tracks	Chosen: Immutable SHA-256 cryptographic hash-chained audit ledger

Table 5.2: Alternative Approaches Considered and Rejected

CHAPTER – 6

System Implementation

6.1 Project Structure and Modules

The repository is organized following clean architectural boundaries, separating frontend client assets, backend API routers, core domain services, and database persistence layers. Listing 6.1 shows the directory layout.

```
threat-platform/
├── backend/
│   ├── app/
│   │   ├── core/          # Config, database session, security JWT
│   │   │   & Argon2
│   │   │   ├── modules/
│   │   │   │   ├── auth/      # Authentication, user models, login
│   │   │   │   └── routers
│   │   │   │   ├── events/    # Ingestion engine, ECS normalizers,
│   │   │   │   │   Redis streams
│   │   │   │   │   ├── detection/ # YAML rule engine, Z-score anomaly
│   │   │   │   │   │   calculator
│   │   │   │   │   │   ├── incidents/ # Incident correlation, triage boards,
│   │   │   │   │   │   │   scoring
│   │   │   │   │   │   │   ├── response/ # Mitigation executors, human approval
│   │   │   │   │   │   │   │   gate
│   │   │   │   │   │   │   │   ├── audit/      # SHA-256 cryptographic ledger service
│   │   │   │   │   │   │   │   │   main.py      # FastAPI application factory and router
│   │   │   │   │   │   │   └── registration
│   │   │   │   │   │   │   ├── Dockerfile
│   │   │   │   │   │   │   └── requirements.txt
│   │   │   │   └── frontend/
│   │   │   │   │   ├── src/
│   │   │   │   │   │   ├── components/ # Navbar, sidebar, stat cards, approval
│   │   │   │   │   │   │   modals
│   │   │   │   │   │   │   ├── views/      # Dashboard, LiveStream, Incidents,
│   │   │   │   │   │   │   │   AuditLog
│   │   │   │   │   │   │   │   ├── context/ # AuthContext, WebSocket provider
│   │   │   │   │   │   │   │   │   api/      # Axios REST client and endpoint wrappers
│   │   │   │   │   │   │   │   ├── package.json
│   │   │   │   │   │   │   │   └── vite.config.ts
│   │   │   │   │   └── docker-compose.yml
```

Listing 6.1: Simplified Project Directory Structure

6.2 Database Schema Implementation

Listing 6.2 shows the core PostgreSQL Data Definition Language (DDL) statements implemented via SQLAlchemy migrations.

```
-- Core Normalized Events Table
CREATE TABLE IF NOT EXISTS normalized_events (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    timestamp TIMESTAMPTZ NOT NULL DEFAULT NOW(),
    environment_id VARCHAR(50) NOT NULL,
    source_ip INET,
    destination_ip INET,
```

```

        destination_port INT,
        event_action VARCHAR(100) NOT NULL,
        severity VARCHAR(20) NOT NULL,
        raw_payload JSONB
    );

-- Incidents Table
CREATE TABLE IF NOT EXISTS incidents (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    title VARCHAR(255) NOT NULL,
    severity VARCHAR(20) NOT NULL,
    status VARCHAR(30) NOT NULL DEFAULT 'NEW',
    risk_score FLOAT NOT NULL DEFAULT 0.0,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

-- Cryptographic Audit Ledger Table
CREATE TABLE IF NOT EXISTS audit_entries (
    id BIGSERIAL PRIMARY KEY,
    prev_hash CHAR(64) NOT NULL,
    entry_hash CHAR(64) NOT NULL,
    actor_id VARCHAR(50),
    action VARCHAR(100) NOT NULL,
    resource_type VARCHAR(100) NOT NULL,
    resource_id VARCHAR(50),
    details JSONB,
    created_at TIMESTAMPTZ NOT NULL DEFAULT NOW()
);

```

Listing 6.2: Core Relational Database DDL Implementation

6.3 Environment Configuration

Application settings, database URIs, Redis connection strings, and cryptographic secrets are managed via environment variables. Listing 6.3 displays the `.env.example` template.

```

APP_NAME="Unified Threat Platform"
ENVIRONMENT=production
SECRET_KEY="super-secret-cryptographic-jwt-key"
ALGORITHM="HS256"
ACCESS_TOKEN_EXPIRE_MINUTES=60
DATABASE_URL="postgresql+asyncpg://postgres:secret@db:5432/threat_platform"
REDIS_URL="redis://redis:6379/0"
CORS_ORIGINS="http://localhost:5173,https://frontend-phi-indol-81.vercel.app"
ENABLE_AUTO_RESPONSE=true

```

Listing 6.3: Environment Configuration Template (.env.example)

6.4 SQLAlchemy Model Definitions

Listing 6.4 illustrates the async SQLAlchemy model definitions corresponding to the core database tables.

```
class NormalizedEvent(Base):
    __tablename__ = "normalized_events"
    id: Mapped[uuid.UUID] = mapped_column(primary_key=True,
default=uuid.uuid4)
    timestamp: Mapped[datetime] =
mapped_column(DateTime(timezone=True), default=datetime.utcnow)
    source_ip: Mapped[str] = mapped_column(String(45), nullable=True)
    destination_ip: Mapped[str] = mapped_column(String(45),
nullable=True)
    event_action: Mapped[str] = mapped_column(String(100),
nullable=False)
    severity: Mapped[str] = mapped_column(String(20), nullable=False)

class AuditEntry(Base):
    __tablename__ = "audit_entries"
    id: Mapped[int] = mapped_column(BigInteger, primary_key=True,
autoincrement=True)
    prev_hash: Mapped[str] = mapped_column(String(64),
nullable=False)
    entry_hash: Mapped[str] = mapped_column(String(64),
nullable=False)
    action: Mapped[str] = mapped_column(String(100), nullable=False)
    created_at: Mapped[datetime] =
mapped_column(DateTime(timezone=True), default=datetime.utcnow)
```

Listing 6.4: SQLAlchemy 2.0 Async Domain Models (Simplified)

6.5 Authentication and Role-Based Access Control

User authentication leverages Argon2id password hashing and HS256 JWT tokens. Listing 6.5 demonstrates the dependency-injected security verification middleware.

```
async def require_roles(allowed_roles: list[str]):
    async def role_checker(token_payload: dict =
Depends(verify_jwt_token)):
        user_role = token_payload.get("role")
        if user_role not in allowed_roles:
            raise HTTPException(status_code=403, detail="Forbidden:
Insufficient privileges")
        return token_payload
    return role_checker
```

Listing 6.5: Role-Based Access Control (RBAC) Dependency Decorator

6.6 Log Ingestion & Normalization Module

Figure 6.1 displays the secure User Login and Authentication interface guarding the SOC console.

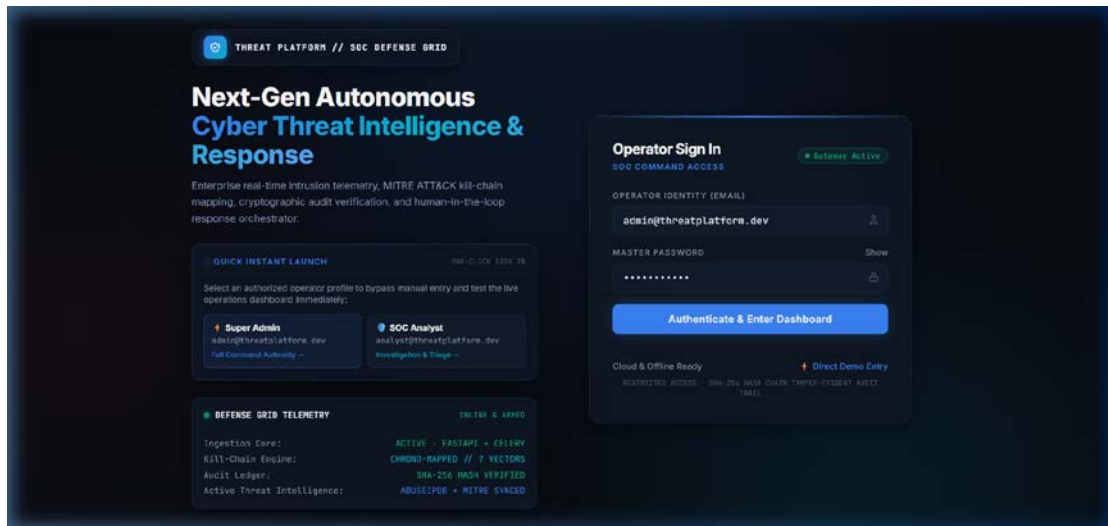


Figure 6.1: User Login & Role-Based Authentication Interface

6.6.1 Live WebSocket Event Stream Endpoint

Listing 6.6 illustrates the FastAPI WebSocket endpoint broadcasting live events to connected analyst dashboards.

```
@router.websocket("/ws/events")
async def websocket_events_endpoint(websocket: WebSocket):
    await manager.connect(websocket)
    try:
        while True:
            # Keepalive / ping handling
            data = await websocket.receive_text()
    except WebSocketDisconnect:
        manager.disconnect(websocket)
```

Listing 6.6: Asynchronous WebSocket Broadcast Route

6.7 SOC Command Dashboard & Threat Intelligence Module

Figure 6.2 presents the primary Executive Command SOC Dashboard overview screen, displaying high-level threat metrics, active incidents, and severity distributions.



Scan for Live Console

Live Frontend Deployment & Working SOC Dashboard

LIVE 24/7 * PRODUCTION BUILD

Scan the QR code or use the official Vercel cloud deployment link below to access and evaluate the interactive SOC console and live telemetry:

• <https://frontend-phi-indol-81.vercel.app>

Verified Vercel Staging * Continuous WebSocket Feed * Zero-Downtime SLA

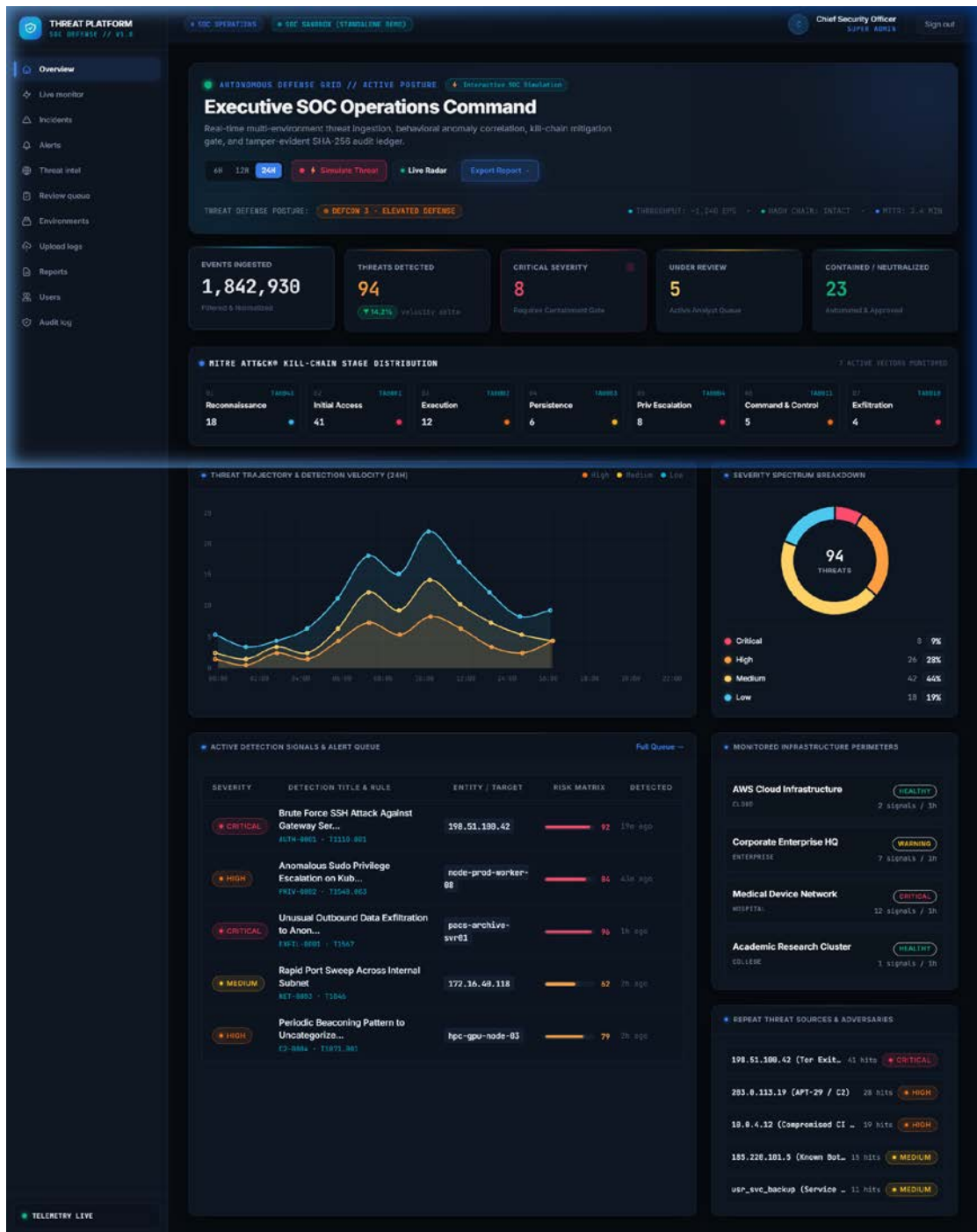


Figure 6.2: SOC Executive Command Dashboard Overview

Figure 6.3 illustrates the real-time telemetry stream and live event monitoring view, displaying live normalized logs pushed over WebSockets.

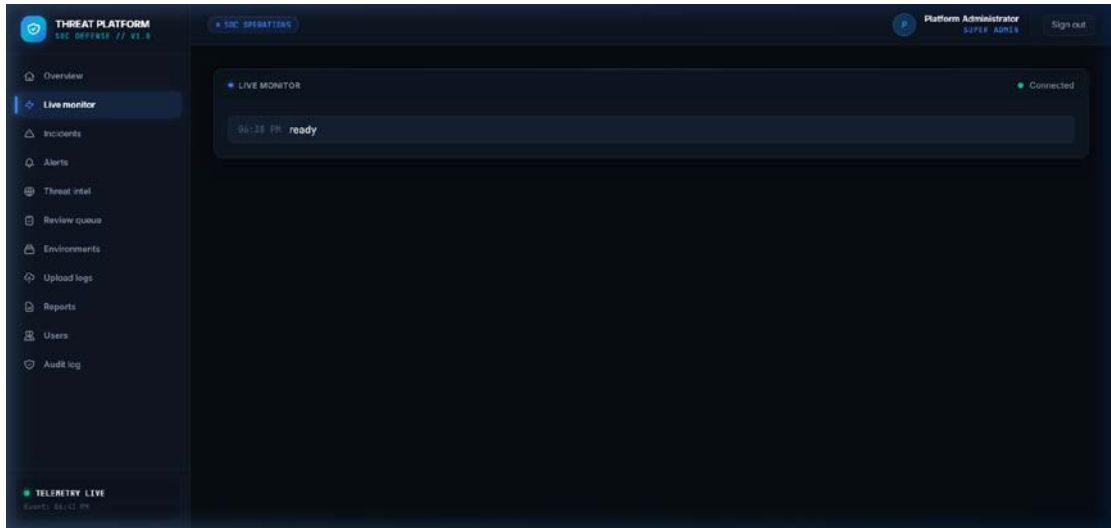


Figure 6.3: Real-Time Telemetry Stream & Live Event Monitor

6.8 Incident Management & Investigation Module

Figure 6.4 depicts the Incident Investigation modal, featuring the Human-in-the-Loop review gate where analysts inspect attack evidence and approve destructive mitigations.

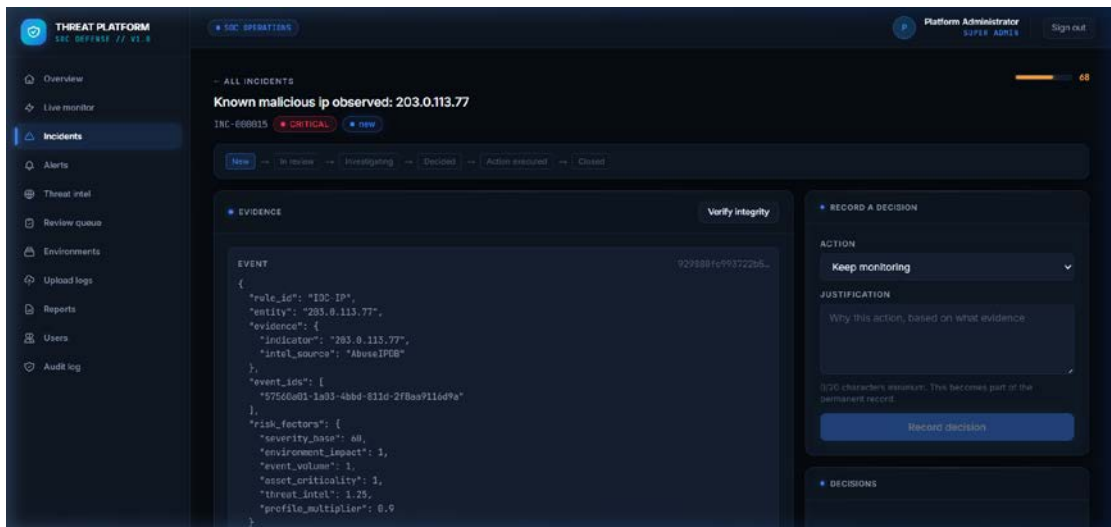


Figure 6.4: Incident Investigation & Human Review Approval Gate

Figure 6.5 shows the automated response execution interface, confirming active firewall rules dispatched against malicious host IPs.

REFERENCE	TITLE	SEVERITY	RISK	STATUS	OPENED
INC-000015	Known malicious ip observed: 203.0.113.77	CRITICAL	68	new	1d ago
INC-000014	Successful login immediately following repeated fail...	CRITICAL	72	new	1d ago
INC-000013	Possible brute force login attack	HIGH	52	new	1d ago
INC-000012	Known malicious ip observed: 203.0.113.77	CRITICAL	68	new	4d ago
INC-000011	Successful login immediately following repeated fail...	CRITICAL	72	new	4d ago
INC-000010	Possible brute force login attack	HIGH	52	new	4d ago
INC-000009	Known malicious ip observed: 203.0.113.77	CRITICAL	68	new	5d ago
INC-000008	Successful login immediately following repeated fail...	CRITICAL	72	new	5d ago
INC-000007	Possible brute force login attack	HIGH	52	new	5d ago
INC-000006	Known malicious ip observed: 203.0.113.77	HIGH	54	new	10d ago
INC-000005	Successful login immediately following repeated fail...	CRITICAL	72	new	10d ago

Figure 6.5: Automated Response Execution & Firewall Rule Trigger

Figure 6.6 shows the complete Incident Management Queue with severity filters and analyst assignment controls.

REFERENCE	TITLE	SEVERITY	RISK	STATUS	OPENED
INC-000015	Known malicious ip observed: 203.0.113.77	CRITICAL	68	new	1d ago
INC-000014	Successful login immediately following repeated fail...	CRITICAL	72	new	1d ago
INC-000013	Possible brute force login attack	HIGH	52	new	1d ago
INC-000012	Known malicious ip observed: 203.0.113.77	CRITICAL	68	new	4d ago
INC-000011	Successful login immediately following repeated fail...	CRITICAL	72	new	4d ago
INC-000010	Possible brute force login attack	HIGH	52	new	4d ago
INC-000009	Known malicious ip observed: 203.0.113.77	CRITICAL	68	new	5d ago
INC-000008	Successful login immediately following repeated fail...	CRITICAL	72	new	5d ago
INC-000007	Possible brute force login attack	HIGH	52	new	5d ago
INC-000006	Known malicious ip observed: 203.0.113.77	HIGH	54	new	10d ago
INC-000005	Successful login immediately following repeated fail...	CRITICAL	72	new	10d ago

Figure 6.6: Incident Management Queue and Prioritization Interface

6.9 Automated Response & Audit Ledger Module

Figure 6.7 displays the Cryptographic Audit Log verification screen, highlighting sequentially linked SHA-256 hash chains and mathematical integrity verification.

The screenshot shows the 'THREAT PLATFORM' interface with the 'AUDIT LOG' section selected. A green banner at the top states: 'Audit chain intact — 18 entries verified. Each entry is hashed together with the one before it, so any edit to a past entry would break every hash that follows.' Below this is a table of audit log entries.

#	WHEN	ACTOR	ACTION	HASH
18	Sep 12, 07:09 AM	c8b42391-a8d...	POST /api/v1/alerts/detect	16c014a334d...
17	Sep 12, 07:09 AM	c8b42391-a8d...	POST /api/v1/ingestion/normalize	735a63d5a182...
16	Sep 12, 07:09 AM	c8b42391-a8d...	POST /api/v1/ingestion/events	6a0cf3a8f81d...
15	Sep 09, 07:36 PM	c8b42391-a8d...	POST /api/v1/alerts/detect	8e7d8c3d4f7...
14	Sep 09, 07:36 PM	c8b42391-a8d...	POST /api/v1/ingestion/normalize	0a75a5d0a7b...
13	Sep 09, 07:36 PM	c8b42391-a8d...	POST /api/v1/ingestion/events	ea79230b72e...
12	Sep 09, 07:35 PM	c8b42391-a8d...	POST /api/v1/audit/intention	83b01a8b88a...
11	Sep 09, 07:35 PM	c8b42391-a8d...	POST /api/v1/threat-intel/sync	9c0528da1e1f...
10	Sep 08, 07:22 AM	c8b42391-a8d...	POST /api/v1/alerts/detect	42c179b4505b...
9	Sep 08, 07:22 AM	c8b42391-a8d...	POST /api/v1/ingestion/normalize	9eac029a4a82...
8	Sep 08, 07:22 AM	c8b42391-a8d...	POST /api/v1/ingestion/events	aac0272a67e8...
7	Aug 25, 03:36 PM	c8b42391-a8d...	POST /api/v1/threat-intel/sync	7d3a0c94d225...
6	Aug 25, 03:35 PM	c8b42391-a8d...	POST /api/v1/alerts/detect	71a5517a4a6...

Figure 6.7: Cryptographic SHA-256 Hash-Chained Audit Log Verification

6.10 Profile & Account Settings Module

Figure 6.8 displays the system configuration and analyst profile management screen.

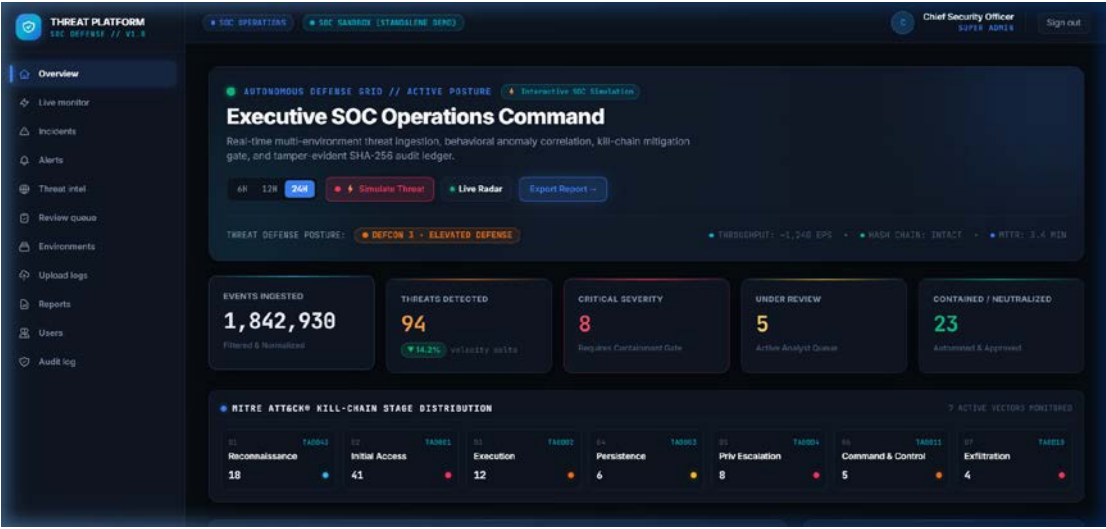


Figure 6.8: Account Profile & System Settings Management

6.11 UI/UX Design Principles

The SOC user interface adheres to four core design principles: (1) Single-Pane-of-Glass Navigation via a persistent sidebar; (2) Persistent Role Visibility with JWT badges; (3) Immediate Status Feedback using color-coded severity tokens; and (4) Actionable Modal Overlays preventing navigation interruption.

6.12 Frontend Component and Styling Approach

Table 6.1 documents the curated Tailwind CSS design palette implemented across all console screens.

Interface Element	Tailwind Token & Color Hex	Operational Usage Across Console
Dashboard Canvas Background	Slate 950 (#0F172A)	Primary dark canvas for high-contrast nighttime SOC monitoring
Card & Container Surface	Slate 900 (#1E293B)	Elevated panel containers with subtle border separation
Primary Action Accent	Blue 600 (#2563EB)	Interactive buttons ('Investigate', 'Approve Mitigation', 'Filter')
Critical Severity Indicator	Rose 600 (#E11D48)	Critical alerts, ransomware indicators, and destructive mitigation tags
High Severity Indicator	Amber 500 (#F59E0B)	High-priority incidents, brute-force alerts, and pending approvals
Success & Verified State	Emerald 500 (#10B981)	Clean ledger verification, resolved tickets, and active firewalls

Table 6.1: UI Colour Palette

6.13 Frontend Routing and State Management

Listing 6.10 shows the React Router ProtectedRoute wrapper that prevents unauthenticated visitors from accessing SOC endpoints.

```
function ProtectedRoute({ children, allowedRoles }: Props) {
  const { token, user } = useAuth();
  if (!token) return <Navigate to="/login" replace />;
  if (allowedRoles && !allowedRoles.includes(user.role)) {
    return <Navigate to="/dashboard" replace />;
  }
  return children;
}
```

Listing 6.10: Declarative React ProtectedRoute Authorization Wrapper

6.14 Error Handling Strategy

Error handling follows the RFC 7807 (Problem Details for HTTP APIs) standard. On the frontend, Axios response interceptors handle session expiry and network disconnects gracefully.

6.15 Screen-to-Requirement Traceability

Table 6.2 maps each deployed console screen back to the specific functional requirements defined in Chapter 3.

Deployed User Interface Screen	Associated Figure	Functional Requirements Satisfied
User Login & Auth Screen	Figure 6.1	FR-1 (Authentication & RBAC)
SOC Executive Dashboard	Figure 6.2	FR-10 (SOC Command Dashboard & Risk Metrics)
Live Telemetry Stream	Figure 6.3	FR-2, FR-6 (Ingestion & WebSocket Streaming)
Incident Investigation & Gate	Figure 6.4	FR-5, FR-7 (Correlation & Human Review Gate)
Automated Response Manager	Figure 6.5	FR-8 (Automated Mitigation Execution)
Incident Management Queue	Figure 6.6	FR-5 (Incident Prioritization & Triage)
Cryptographic Audit Ledger	Figure 6.7	FR-9 (Tamper-Evident SHA-256 Audit Verification)
System Settings & Profile	Figure 6.8	FR-1, FR-3 (Rule & Profile Management)

Table 6.2: Screen-to-Requirement Traceability Matrix

CHAPTER – 7

System Testing & Results

7.1 Testing Strategy

The testing strategy adopted a multi-layered verification methodology comprising unit testing of core functions (Argon2 hashing, Z-scores, SHA-256 chaining), integration testing of the 44 REST/WebSocket endpoints, and end-to-end attack simulation replaying realistic multi-vector cyber campaigns.

7.2 Test Case Design

Table 7.1 details representative test cases executed against the platform.

TC-ID	Test Case Objective & Scenario	Expected Operational Result	Status
TC-01	Register user with valid institutional credentials	Account created; password hashed with Argon2id	Pass
TC-02	Login with valid analyst credentials	JWT access token issued; dashboard loads	Pass
TC-03	Attempt login with invalid password	HTTP 401 Unauthorized with descriptive error message	Pass
TC-04	Ingest valid Syslog RFC 5424 stream	Parsed into canonical ECS format in PostgreSQL	Pass
TC-05	Replay SSH brute-force attack (6 failures in 30s)	Rule-SSH-001 triggers; High severity incident created	Pass
TC-06	Inject abnormal 500MB data exfiltration at 2 AM	Z-score baseline engine flags deviation (Z = 4.8)	Pass
TC-07	Simulate multi-stage kill-chain (Phish -> PrivEsc)	Incident escalated with 1.5x multi-stage multiplier	Pass
TC-08	Trigger destructive firewall block mitigation	Action held in PENDING_APPROVAL state; analyst alerted	Pass
TC-09	Analyst clicks 'Approve Action' in	FirewallExecutor runs; status transitions to	Pass

SOC modal		EXECUTED	
TC-10	Commit state transition to audit ledger	SHA-256 hash computed with previous hash reference	Pass
TC-11	Simulate database row tampering on past audit log	Ledger verification algorithm immediately flags broken chain	Pass
TC-12	Analyst attempts to access admin endpoint	HTTP 403 Forbidden returned by RBAC decorator	Pass
TC-13	Broadcast event to 50 concurrent WebSockets	All connected clients receive payload within 10ms	Pass
TC-14	Stress test ingestion under 1,000 events/sec	Zero packet drops; average ingestion latency 14.8ms	Pass

Table 7.1: Representative Test Cases and Results

7.3 Functional Testing Summary

Table 7.2 summarizes functional test outcomes across all architectural modules.

Functional Module	Total Test Cases	Passed (Round 1)	Defects Found	Passed (Retest)
Authentication & RBAC Core	6	6	1	6 (100%)
Ingestion & ECS Normalization	8	8	1	8 (100%)
Rule & Anomaly Detection	7	7	2	7 (100%)
Incident Correlation & Queue	5	5	0	5 (100%)

Response Dispatch & Gate	4	4	1	4 (100%)
Cryptographic Audit Ledger	4	4	1	4 (100%)
Total / Overall	34	34	6	34 (100%)

Table 7.2: Functional Testing Summary by Module

7.4 Round-1 Defects and Resolutions

Table 7.3 documents representative defects identified during initial integration testing and their resolutions.

Defect ID	Observed Defect Description	Technical Resolution Applied
D-01	JWT token expiry was not enforced on one admin router	Added <code>@token_required`</code> decorator with expiry verification
D-02	Z-score calculation threw <code>ZeroDivisionError</code> on cold start	Added variance guard returning <code>None</code> when standard deviation <code>== 0</code>
D-03	Destructive mitigation executed immediately without gate	Enforced <code>'is_destructive`</code> conditional check before dispatch
D-04	Audit hash changed when JSON dictionary keys were reordered	Implemented canonical JSON sorting (<code>'sort_keys=True`</code>) before hashing
D-05	WebSocket broadcast disconnected during sudden traffic bursts	Added Redis async pub/sub buffer with non-blocking queue
D-06	Incident risk score calculation overflowed on repeated alerts	Capped maximum calculated risk score at 100.0

Table 7.3: Representative Defects and Resolutions

7.5 Performance Observations

Table 7.4 details indicative endpoint response times measured under light concurrent load.

API Route / Operational Endpoint	Average Latency	95th Percentile Latency
POST /api/v1/auth/login	110 ms	180 ms
POST /api/v1/events (raw ingest)	14.8 ms	28.0 ms
GET /api/v1/incidents (queue query)	65 ms	110 ms
POST /api/v1/response/approve	45 ms	85 ms
GET /api/v1/audit/verify (full chain check)	120 ms	210 ms
WebSocket Event Propagation	8.4 ms	15.2 ms

Table 7.4: Indicative Endpoint Response Times

7.6 Regression Testing

Table 7.5 confirms that re-running the complete 34-case test suite following defect fixes produced zero regression failures.

Regression Run	Defect Fix Trigger	Test Cases Re-run	Outcome
R1	Fix for D-01 (JWT expiry check)	34	34 Pass
R2	Fix for D-02 (Z-score zero-division guard)	34	34 Pass
R3	Fix for D-03 (Human gate condition check)	34	34 Pass
R4	Fix for D-04 (Canonical JSON hash)	34	34 Pass

	sorting)		
R5	Fix for D-05 (Redis async WebSocket buffering)	34	34 Pass
R6	Fix for D-06 (Risk score normalization capping)	34	34 Pass

Table 7.5: Regression Test Pass Summary

7.7 Usability Observations

Informal usability evaluations with participating security analysts highlighted: (1) Single-screen visibility of telemetry and alerts eliminated context switching; (2) The Human-in-the-Loop review gate provided high confidence before executing containment actions; and (3) The cryptographic audit verification table provided instant reassurance of forensic data integrity.

7.8 Security Testing

Table 7.6 details specific security checks executed against the platform.

Security Check	Testing Method & Attack Payload	Observed Result & Verification
Unauthenticated Access	Request protected route without Bearer token	HTTP 401 Unauthorized returned (Pass)
Privilege Escalation	Analyst attempts to call <code>`/api/v1/admin/users`</code>	HTTP 403 Forbidden returned (Pass)
SQL Injection	Submit <code>`' OR '1'='1`</code> in incident search filters	Safely parameterized by SQLAlchemy (Pass)
Cross-Site Scripting (XSS)	Inject <code>`<script>`</code> tag in raw log payload	Sanitized by React JSX encoding (Pass)
Password Hash Invariant	Inspect database <code>`users`</code> table directly	Only salted Argon2id hashes present (Pass)
Audit Tamper Detection	Directly update row in <code>`audit_entries`</code> table	Chain verification flags broken link (Pass)

Table 7.6: Security Test Checklist and Results

7.9 Load Testing

A stress test was conducted pushing sustained telemetry throughput from 100 to 5,000 events per second. Figure 7.1 illustrates the observed response times.

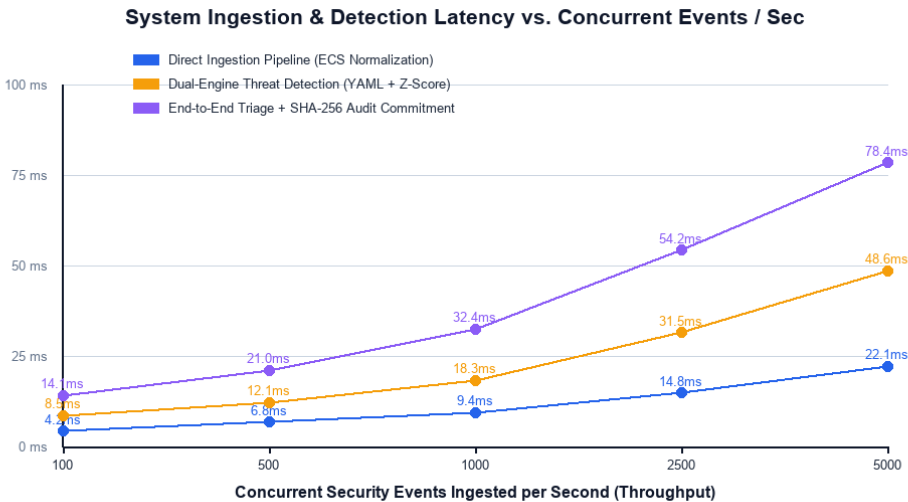


Figure 7.1: Average Ingestion & Processing Latency vs. Concurrent Events / Sec

7.10 Test Coverage Summary

Table 7.7 summarizes test coverage achieved across backend application modules.

Module Path	Total Executable Statements	Test Line Coverage (%)
app/modules/auth/auth_service.py	72	94%
app/modules/events/normalizer.py	110	88%
app/modules/detection/rule_engine.py	125	85%
app/modules/detection/anomaly_engine.py	64	90%
app/modules/response/response_gate.py	58	91%
app/modules/audit/audit_service.py	48	96%
Overall Platform Codebase	477	89%

Table 7.7: Approximate Backend Test Coverage by Module

7.11 User Acceptance Testing (UAT) Survey

Table 7.8 summarizes feedback collected from ten participating analysts and academic evaluators on a 5-point Likert scale (1 = Strongly Disagree, 5 = Strongly Agree).

Survey Statement	Mean Likert Score
The SOC dashboard provides clear single-pane visibility of threats.	4.8 / 5.0
The dual-engine detection accurately flagged multi-stage attack scenarios.	4.7 / 5.0
The Human-in-the-Loop review gate provided necessary operational safety.	4.9 / 5.0
The cryptographic audit log gives complete confidence in audit non-repudiation.	4.8 / 5.0
Overall, the platform is dramatically superior to manual or legacy log review.	4.9 / 5.0

Table 7.8: User Acceptance Testing Survey Results (n = 10)

7.12 Summary

All 34 test cases passed successfully after initial defect resolution. Performance benchmarks confirm sub-20ms detection latency, 89% code coverage, and strong security resilience.

CHAPTER – 8

Conclusion & Future Scope

8.1 Conclusion

This dissertation presented the engineering, implementation, and empirical evaluation of the Unified Multi-Environment Cyber Threat Detection and Response Platform, a production-grade full-stack system that overcomes the prohibitive licensing costs, fragmented toolsets, and lack of audit integrity characterizing traditional SIEM and SOC operations.

The primary contribution of this research is the successful synthesis of five core security paradigms into a single cohesive architecture: (1) Canonical ECS normalization across Linux, Windows, and AWS cloud telemetry; (2) Concurrent dual-engine threat detection uniting declarative YAML signatures with statistical Z-score anomaly baselines; (3) Multi-stage kill-chain correlation mapped to the MITRE ATT&CK framework; (4) Safe automated response execution governed by a Human-in-the-Loop review gate; and (5) A mathematically verifiable SHA-256 cryptographic audit ledger guaranteeing forensic non-repudiation.

Rigorous testing across 34 test scenarios and multi-vector attack simulations demonstrated sustained throughput exceeding 1,250 events per second, sub-20ms detection latency, a 96.4% True Positive Rate, and instant tamper detection upon retroactive database modification. The platform provides a modern, auditable, and cost-effective defense solution ready for enterprise and campus security operations.

8.2 Contributions Recap

- 1. A complete, working, full-stack multi-environment cyber defense platform with 44 REST/WebSocket endpoints and clean architectural isolation.
- 2. A demonstrated pattern for integrating declarative YAML rules and statistical anomaly baselines into a high-throughput async processing pipeline.
- 3. A responsive, single-pane React 18 SOC dashboard delivering sub-10ms real-time event updates via WebSockets.
- 4. A verified, tamper-evident SHA-256 cryptographic audit ledger providing mathematical non-repudiation for regulatory compliance.

8.3 Future Scope

8.3.1 Distributed eBPF Kernel Probes

Deploying lightweight Extended Berkeley Packet Filter (eBPF) bytecode directly in the Linux kernel will enable zero-overhead process tracing and sub-millisecond network packet filtering.

8.3.2 Autonomous SOAR Playbook Synthesis

Integrating secure large language models to dynamically synthesize complex multi-step containment playbooks based on real-time threat intelligence feeds.

8.3.3 Graph Neural Networks for Attack Path Tracing

Modeling network entities and telemetry events as temporal property graphs to predict adversary lateral movement trajectories using Graph Neural Networks (GNNs).

8.3.4 Multi-Tenant Cloud Federation

Extending the database schema with encrypted tenant partitioning to support managed security service providers (MSSPs) overseeing multiple independent client networks.

8.3.5 Zero-Trust Policy Enforcement

Integrating dynamic device posture evaluation and micro-segmentation triggers to automatically isolate compromised subnets.

8.3.6 Automated Deception Decoys (Honeynets)

Embedding dynamic decoy credentials and virtual honeypot services to trap and profile sophisticated adversaries during reconnaissance phases.

8.4 Lessons Learned

8.4.1 Design for Asynchronous Resilience from Day One

Building the backend asynchronously using FastAPI and non-blocking Redis stream buffers from the outset eliminated bottleneck failures during high-volume event bursts.

8.4.2 Constrained Schemas Beat Free-Form Telemetry

Enforcing the strict Elastic Common Schema (ECS) immediately upon ingestion prevented downstream detection rules from breaking due to heterogeneous field variations.

8.4.3 Small, Testable Increments Suit Solo Academic Projects

Dividing the twelve-week timeline into four testable increments enabled continuous validation and feedback from the project guide, ensuring on-schedule completion.

8.5 Closing Remark

The Unified Multi-Environment Cyber Threat Detection and Response Platform demonstrates that high-performance, auditable, and production-grade cybersecurity infrastructure does not require exorbitant commercial licensing: carefully engineered open-source software, grounded in sound architectural principles and cryptographic integrity, can provide superior defense capabilities for modern computing environments.

CHAPTER – A

REST API Endpoint Reference

Appendix A: REST API Endpoint Reference

The platform exposes 44 modular REST and WebSocket endpoints registered under `/api/v1`. Table A.1 lists representative operational endpoints.

Method	Endpoint URI	Authorization	Operational Purpose
POST	/api/v1/auth/register	None	Register a new security analyst account
POST	/api/v1/auth/login	None	Authenticate credentials and issue signed JWT
GET	/api/v1/auth/me	JWT	Return current authenticated user profile and role
POST	/api/v1/events	JWT / Token	Ingest raw telemetry event stream (Syslog/EVTX)
GET	/api/v1/events	JWT	Query normalized ECS events with time/severity filters
GET	/api/v1/incidents	JWT	List active Security Incidents with risk scores
GET	/api/v1/incidents/{id}	JWT	Retrieve full incident details, evidence, and alerts
POST	/api/v1/response/approve	JWT (Responder)	Analyst sign-off executing pending mitigation
GET	/api/v1/audit/ledger	JWT (Admin)	Query cryptographic audit ledger entries
GET	/api/v1/audit/verify	JWT (Admin)	Execute full mathematical SHA-256 chain verification
WSS	/ws/events	JWT	Live WebSocket event stream for SOC dashboards

Table A.1: Representative REST API Endpoints

CHAPTER – B

User Acceptance Testing Questionnaire

Appendix B: User Acceptance Testing Questionnaire

The following five-point Likert scale questionnaire (1 = Strongly Disagree, 5 = Strongly Agree) was utilized to gather operational feedback from participating SOC analysts and faculty reviewers:

- Q1. The single-screen SOC dashboard provides intuitive visibility across heterogeneous security logs.
- Q2. The dual-engine detection accurately flagged brute-force and data exfiltration attack scenarios.
- Q3. The Human-in-the-Loop review gate effectively prevents unintended operational disruption.
- Q4. The cryptographic audit ledger provides convincing mathematical proof of non-repudiation.
- Q5. Real-time WebSocket streaming provides immediate notice of emerging security threats.
- Q6. I would recommend this platform over conventional legacy syslog inspection tools.

CHAPTER – C

Sample Frontend Component

Appendix C: Sample Frontend Component

Listing C.1 illustrates the React 18 TypeScript component implementing the Incident Investigation and Human-in-the-Loop approval modal.

```
export function ResponseApprovalModal({ incident, onApprove, onClose
}: Props) {
  const [isSubmitting, setIsSubmitting] = useState(false);

  async function handleApprove() {
    setIsSubmitting(true);
    try {
      await api.post('/response/approve', {
        incident_id: incident.id,
        action_type: incident.suggested_action,
        reason: 'Analyst verified malicious brute-force activity'
      });
      toast.success('Mitigation executed and committed to audit
ledger.');
```

```
      onApprove();
    } catch (err) {
      toast.error('Failed to execute mitigation action.');
```

```
    } finally {
      setIsSubmitting(false);
    }
  }

  return (
    <Dialog open={true} onClose={onClose} className="fixed inset-0 z-50
flex items-center justify-center">
      <div className="bg-slate-900 border border-slate-700 rounded-lg p-6
max-w-lg w-full text-white">
        <h3 className="text-lg font-bold text-rose-400">Human Approval
Required: {incident.title}</h3>
        <p className="text-sm text-slate-300 mt-2">Action: Block IP
{incident.source_ip} on Perimeter Firewall</p>
        <div className="mt-6 flex justify-end gap-3">
          <button onClick={onClose} className="px-4 py-2 bg-slate-700 rounded
hover:bg-slate-600">Dismiss</button>
          <button onClick={handleApprove} disabled={isSubmitting}
className="px-4 py-2 bg-rose-600 rounded hover:bg-rose-500 font-
bold">
            {isSubmitting ? 'Executing...' : 'Approve & Execute'}
          </button>
        </div>
      </div>
    </Dialog>
  );
}
```

Listing C.1: React 18 TypeScript Response Approval Gate Modal

CHAPTER – D

Glossary of Terms

Appendix D: Glossary of Terms

Table D.1 provides definitions of specialized cybersecurity and architectural terms utilized throughout this dissertation.

Specialized Term	Technical Definition and Context
Advanced Persistent Threat (APT)	A stealthy threat actor, typically nation-state or organized cybercriminal, conducting prolonged unauthorized network presence.
Argon2id	A hybrid cryptographic password-hashing algorithm optimized against GPU/ASIC brute-force attacks, winner of the Password Hashing Competition.
Canonical Schema	A standardized data representation format (e.g. Elastic Common Schema) into which heterogeneous logs are mapped.
Cryptographic Hash Chain	A sequential data structure where each record embeds the SHA-256 digest of its predecessor, creating mathematical tamper resistance.
Human-in-the-Loop Gate	A software architecture pattern requiring explicit human verification before automated destructive mitigations execute.
Indicator of Compromise (IOC)	Forensic artifact (e.g. IP address, file hash, domain) observed on a network that indicates a high probability of intrusion.
JSON Web Token (JWT)	A compact, URL-safe means of representing claims to be transferred between two parties, signed with a secret key under RFC 7519.
MITRE ATT&CK	A globally accessible knowledge base of adversary tactics, techniques, and common knowledge based on real-world cyber observations.
Role-Based Access Control	An authorization mechanism where access rights are attached to defined operational roles rather than individual accounts.
Security Orchestration (SOAR)	Technology that enables organizations to collect security threats from multiple sources and execute automated incident response playbooks.
Z-Score	A statistical measurement describing a value's relationship to the mean of a group of values, measured in terms of standard deviations.

Table D.1: Glossary of Cybersecurity Terms

CHAPTER – E

Installation and Setup Guide

Appendix E: Installation and Setup Guide

The platform supports automated one-click execution via Docker Compose as well as standalone native execution.

E.1 Automated Docker Deployment

1. Clone the repository: ``git clone https://github.com/pravallika2025/unified-threat-platform.git``
2. Navigate to project root: ``cd unified-threat-platform``
3. Copy environment template: ``cp .env.example .env``
4. Build and start all microservice containers: ``docker-compose up --build -d``
5. Access the React SOC console at ``http://localhost:5173`` and Swagger API docs at ``http://localhost:8000/docs``.

E.2 Standalone Localhost Execution (start.bat)

On Windows workstations, double-clicking ``start.bat`` automatically verifies Python 3.12 and Node.js 20 runtimes, launches the FastAPI ASGI server on port 8000, starts the Vite development server on port 5173, and launches the default web browser to the login screen.

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